

On the stability of scale-space metrics

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Abstract

We study the stability of a classical family of metrics defined over functions’ Gaussian scale-space representations, focusing on the comparison of images (functions of two variables). These metrics have precedents both in harmonic analysis, specifically the theory of Besov spaces, and in classical methods of image processing; special cases are also known to be metrically equivalent to certain Wasserstein distances. We quantify these metrics’ robustness to geometric deformations, and introduce rotationally-invariant versions that are stable to changes in angle when comparing tomographic projections. We also describe computationally efficient algorithms for evaluating the metrics from finite samples, and prove their robustness to additive noise. The results are illustrated through numerical experiments.

1 Introduction

This paper studies a family of metrics between real-variable functions, exploring their stability to geometric deformations and additive noise. The definitions of these metrics makes use of a function’s “scale-space” representation, in which the function is convolved with a sequence of Gaussian kernels of varying widths. Each function is thereby replaced by a stack of functions, of varying resolution. The distance $D_{\alpha,p,r}(f,g)$ between two functions f and g in $L^r \cap L^1$ is then defined as a weighted sum of the distances across each stack; the precise definition is given in Section 2.

Such metrics, and the ideas underlying their construction, have previously appeared in numerous works:

- In the harmonic analysis literature, the norms that induce these metrics are equivalent to the dual norms of Besov spaces (certain classes of smooth functions) [46, 41].
- Classical imaging processing techniques make use of an image’s scale-space representation via Gaussian smoothing [26]. For instance, features extracted from the scale-space representation of a function are used in the well-known SIFT algorithm for feature detection [28, 27].
- The special case where $p = r = 1$ and $0 < \alpha < 1$ is known to be metrically equivalent to the Wasserstein distance with ground distance $|x - y|^\alpha$; see [23], as well as the related works [43, 10], which employ multiscale analysis via wavelets rather than Gaussians.
- Analogous metrics over more abstract spaces that exhibit a multiscale structure (such as spaces endowed with hierarchical tree metrics) have been studied in data science applications; for instance, see [21, 23, 20, 31, 32, 25, 9].

In the present work, we explore the stability of these metrics $D_{\alpha,p,r}$ to distortions of their inputs, both under measure-preserving geometric deformations and additive noise. We focus specifically on applications to image-processing (i.e. to functions on $\mathbb{R}^2$, and their discretizations), though many of our results may be generalized to other dimensions. One of the scientific motivations for our work is the robust comparison of 2D tomographic projections of 3D functions, a setting which occurs in applications like cryo-electron microscopy (cryo-EM) [5, 44, 47, 11, 51]. Various metrics related to Wasserstein distances have been proposed for image comparison in cryo-EM recently [42, 38], not to mention scores of other applications and theoretical results [39, 35, 7, 43, 24, 49, 50, 40, 34]. Because the metrics $D_{\alpha,p,r}$ generalize a metric equivalent to Wasserstein, it is natural to explore their properties as well.

Because the images in cryo-EM typically undergo random in-plane rotations, it is important to define a rotation-invariant version of the metric $D_{\alpha,p,r}$, which minimizes the distance between the input functions over all in-plane rotations, as is done in [42] and [38]. As will become clear later, for this to be computationally feasible we restrict $r = 2$; see Section 5.2 for details.

The main contributions of the paper are as follows:

- We bound the distortion between $D_{\alpha,p,r}(f,g)$ when g is a perturbation of f , in terms of the maximum displacement of the perturbation. (In fact, the bound extends to the more general setting where f and g are projections of functions that are perturbations of each other.)

- If f and g are 2D projections of a fixed 3D function, we prove robustness of the rotationally-invariant metric $D_{\alpha,p,r}(f,g; \text{SO}(2))$ to changes in the projection angle.
- When $r = 2$, we define a fast algorithm for calculating both $D_{\alpha,p}(f,g)$ and the rotationally-invariant metric $D_{\alpha,p}(f,g; \text{SO}(2))$. For functions discretized on an n -by- n Cartesian grid, the cost of these algorithms scale like $O(n^2 \log(n))$.
- We prove robustness of the discretization to additive sub-Gaussian noise.
- Finally, we report on numerical experiments illustrating the aforementioned results.

2 The scale-space metrics

The metrics will be defined by several user-set parameters: $\tau_0 > 0$, $0 < \omega < 1$, $r \geq 1$, $p \geq 1$, and $\alpha > 0$. Typically, we will take $\omega = 1/2$; and when rotational-invariance is needed, $r = 2$.

For a value $\tau > 0$, denote by G_τ the Gaussian kernel at scale τ , defined by

$$G_\tau(x) = \frac{1}{(\pi\tau)^{d/2}} e^{-|x|^2/\tau}. \quad (1)$$

For fixed values $\tau_0 > 0$ and $0 < \omega < 1$, and for $k \geq 0$, define

$$\tau_k = \tau_0 \omega^k. \quad (2)$$

Fix $\alpha > 0$, $p \geq 1$, and $r \geq 1$. For a function f in $L^r(\mathbb{R}^d)$, we define the norm

$$\|f\|_{\Gamma_{p,r}^\alpha} = \left(\sum_{k \geq 0} \tau_k^{p\alpha/2} \|G_{\tau_k} * f\|_{L^r}^p \right)^{1/p}, \quad (3)$$

and denote the induced distance between functions f and g as follows:

$$D_{\alpha,p,r}(f,g) = \|f - g\|_{\Gamma_{p,r}^\alpha}. \quad (4)$$

When $r = 2$, we will denote this more simply by $D_{\alpha,p}(f,g)$; and when $r = 2$ and $p = 1$, we will denote it by $D_\alpha(f,g)$.

Given a subgroup $\mathcal{G} \leq O(d)$, we define the $\mathcal{G}$ -invariant distance between f and g by

$$D_{\alpha,p,r}(f,g; \mathcal{G}) = \min_{U \in \mathcal{G}} \|f - g \circ U\|_{\Gamma_{p,r}^\alpha}. \quad (5)$$

$D_{\alpha,p,r}(f,g; \mathcal{G})$ is really a distance between the orbits of f and g under the action of $\mathcal{G}$. In this work, we will only consider the case where $\mathcal{G} = \text{SO}(2)$, for comparing functions of two variables.

2.1 Difference-of-Gaussians

It is known that the norm $\|f\|_{\Gamma_{p,r}^\alpha}$ is in fact equivalent (in the sense that the ratios are bounded above and below by constants independent of f) to the norm

$$\|f\|_{\Gamma_{p,r}^\alpha}^{(1)} = \left(\tau_0^{p\alpha/2} \|G_{\tau_0} * f\|_{L^r}^p + \sum_{k \geq 0} \tau_k^{p\alpha/2} \|(G_{\tau_k} - G_{\tau_{k+1}}) * f\|_{L^r}^p \right)^{1/p}. \quad (6)$$

More precisely, we have the following result:

Proposition 2.1. *There is a universal constant $C > 0$ such that for all f in L^r ,*

$$\frac{1}{C} \cdot \|f\|_{\Gamma_{p,r}^\alpha}^{(1)} \leq \|f\|_{\Gamma_{p,r}^\alpha} \leq \frac{C}{1 - \omega^{\alpha/2}} \cdot \|f\|_{\Gamma_{p,r}^\alpha}^{(1)}. \quad (7)$$

Remark 1. Results of this form are well known in classical Besov space theory; indeed, there are many equivalent formulas for Besov norms and their duals [46, 41]. Though there is nothing essentially novel in the result, for the reader's convenience, we give a self-contained, elementary proof.

Remark 2. The norm $\|f\|_{\Gamma_{p,r}^\alpha}^{(1)}$ is of interest in that it convolves with the difference-of-Gaussian filters $G_{\tau_k} - G_{\tau_{k+1}}$; that is, it applies a sequence of band-pass filters to f , rather than low-pass filters. The difference-of-Gaussians filters are another classical tool of image processing, useful for edge detection [30, 52]. Related ideas underly the Laplacian pyramids algorithm [8], among others.

Remark 3. Note that upper bound on $\|f\|_{\Gamma_{p,r}^\alpha}/\|f\|_{\Gamma_{p,r}^\alpha}^{(1)}$ grows to infinity as $\alpha \rightarrow 0^+$, whereas the lower bound is a universal constant. This behavior is to be expected, and understanding why may give some intuition as to the way the norms are defined. Reasoning informally, each term $G_{\tau_k} * f$ appearing in $\|f\|_{\Gamma_{p,r}^\alpha}$ serves as a low-pass filter, which essentially preserves the frequencies ξ in the range $0 \leq |\xi| \leq A/\sqrt{\tau_k}$ (for some $A > 0$, with leakage, that we will ignore, due to the non-compact support of $\widehat{G_{\tau_k}}$). On the other hand, each term $(G_{\tau_k} - G_{\tau_{k+1}}) * f$ appearing in $\|f\|_{\Gamma_{p,r}^\alpha}^{(1)}$ serves as a band-pass filter, preserving the frequencies ξ of f in the range $A/\sqrt{\tau_k} < |\xi| < A/\sqrt{\tau_{k+1}}$ (again, ignoring leakage outside this range). In other words, each summand of $\|f\|_{\Gamma_{p,r}^\alpha}^{(1)}$ effectively measures f 's frequency content in the *disjoint* regions $A/\sqrt{\tau_k} < |\xi| < A/\sqrt{\tau_{k+1}}$, whereas each summand of $\|f\|_{\Gamma_{p,r}^\alpha}$ measures the frequency content in the *overlapping* regions $0 \leq |\xi| < A/\sqrt{\tau_k}$; the inequality $\|f\|_{\Gamma_{p,r}^\alpha}^{(1)} \leq C \cdot \|f\|_{\Gamma_{p,r}^\alpha}$ is therefore not surprising, since the series defining $\|f\|_{\Gamma_{p,r}^\alpha}$ adds up many redundant contributions from overlapping regions. On the other hand, for each k , though the terms $\|G_{\tau_\ell} * f\|_{L^r}^p$, $\ell \geq k$, all capture frequencies from the region $A/\sqrt{\tau_k} < |\xi| < A/\sqrt{\tau_{k+1}}$, the geometrically decreasing weights $\tau_\ell^{p\alpha/2}$ ensure that the weighted sum of all these contributions is still proportional to the first term $\|G_{\tau_k} * f\|_{L^r}^p$, with a constant that grows as the weights $\tau_\ell^{p\alpha/2}$ decay slower, i.e. as $\alpha \rightarrow 0^+$.

It will first be convenient to state an elementary lemma, which we prove for the reader's convenience.

Lemma 2.2. *Let $0 < \delta < 1$. Then*

$$\left(\frac{1}{1-\delta^p}\right)^{1/p} \leq \frac{1}{1-\delta} \quad (8)$$

for all $p \geq 1$.

Proof of Lemma 2.2. The inequality is equivalent to

$$1 - \delta^p - (1 - \delta)^p \geq 0. \quad (9)$$

Equality holds when $p = 1$, and since $0 < \delta < 1$, the right side is increasing as a function of p (since both δ^p and $(1 - \delta)^p$ decrease as p grows), proving the result. $\square$

Proof of Proposition 2.1. First, using the inequality

$$\|(G_{\tau_{k+1}} - G_{\tau_k}) * f\|_{L^r}^p \leq 2^p (\|G_{\tau_{k+1}} * f\|_{L^r}^p + \|G_{\tau_k} * f\|_{L^r}^p) \quad (10)$$

it immediately follows that

$$\|f\|_{\Gamma_{p,r}^\alpha}^{(1)} \leq C \cdot \|f\|_{\Gamma_{p,r}^\alpha}, \quad (11)$$

for a universal constant $C > 0$.

For the reverse direction, from the telescoping sum

$$G_{\tau_k} * f = G_{\tau_0} * f + \sum_{\ell=1}^k (G_{\tau_\ell} - G_{\tau_{\ell-1}}) * f \quad (12)$$

it follows that

$$\|G_{\tau_k} * f\|_{L^r} \leq \|G_{\tau_0} * f\|_{L^r} + \sum_{\ell=1}^k \|(G_{\tau_\ell} - G_{\tau_{\ell-1}}) * f\|_{L^r}. \quad (13)$$

Consequently,

$$\begin{aligned} \tau_k^{\alpha/2} \|G_{\tau_k} * f\|_{L^r} &\leq \tau_k^{\alpha/2} \|G_{\tau_0} * f\|_{L^r} + \sum_{\ell=1}^k \tau_k^{\alpha/2} \tau_\ell^{-\alpha/2} \tau_\ell^{\alpha/2} \|(G_{\tau_\ell} - G_{\tau_{\ell-1}}) * f\|_{L^r} \\ &= \tau_k^{\alpha/2} \|G_{\tau_0} * f\|_{L^r} + \sum_{\ell=1}^k \omega^{(k-\ell)\alpha/2} \cdot \tau_\ell^{\alpha/2} \|(G_{\tau_\ell} - G_{\tau_{\ell-1}}) * f\|_{L^r}. \end{aligned} \quad (14)$$

Define $A[m] = \omega^{m\alpha/2}$ for $m \geq 0$, and $A[m] = 0$ for $m < 0$; and define $B[\ell] = \tau_\ell^{\alpha/2} \|(G_{\tau_\ell} - G_{\tau_{\ell-1}}) * f\|_{L^r}$ for $\ell > 0$, and $B[\ell] = 0$ for $\ell \leq 0$. Then the series on the right side of (14) may be written as a convolution between A and B over $\mathbb{Z}$:

$$\sum_{\ell=1}^k \omega^{(k-\ell)\alpha/2} \cdot \tau_\ell^{\alpha/2} \|(G_{\tau_\ell} - G_{\tau_{\ell-1}}) * f\|_{L^r} = \sum_{\ell=-\infty}^{\infty} A[k-\ell]B[\ell] = (A * B)[k]. \quad (15)$$

Note that $(A * B)[k] = 0$ if $k \leq 0$, since in that case all the terms $A[k - \ell] = 0$ for $\ell > 0$, and $B[\ell] = 0$ for $\ell \leq 0$. From Young's inequality (see, for instance, Theorem 1.2.12 in [16]), summing over k in (14) gives the bound

$$\begin{aligned}
\left(\sum_{k=0}^{\infty} \left(\sum_{\ell=1}^k \tau_k^{\alpha/2} \tau_\ell^{-\alpha/2} \tau_\ell^{\alpha/2} \| (G_{\tau_\ell} - G_{\tau_{\ell-1}}) * f \|_{L^r} \right)^p \right)^{1/p} &\leq \left(\sum_{k=0}^{\infty} |(A * B)[k]|^p \right)^{1/p} \\
&\leq \left(\sum_{k=-\infty}^{\infty} A[k] \right) \cdot \left(\sum_{\ell=-\infty}^{\infty} B[\ell]^p \right)^{1/p} \\
&= \left(\sum_{k=0}^{\infty} \omega^{k\alpha/2} \right) \cdot \left(\sum_{\ell=1}^{\infty} \tau_\ell^{p\alpha/2} \| (G_{\tau_\ell} - G_{\tau_{\ell-1}}) * f \|_{L^r}^p \right)^{1/p} \\
&= \left(\frac{1}{1 - \omega^{\alpha/2}} \right) \cdot \left(\sum_{\ell=1}^{\infty} \tau_\ell^{p\alpha/2} \| (G_{\tau_\ell} - G_{\tau_{\ell-1}}) * f \|_{L^r}^p \right)^{1/p} \quad (16)
\end{aligned}$$

Consequently, since

$$\|G_{\tau_k} * f\|_{L^r} \leq \|G_{\tau_0} * f\|_{L^r} + \sum_{\ell=1}^k \|(G_{\tau_\ell} - G_{\tau_{\ell-1}}) * f\|_{L^r}, \quad (17)$$

and hence

$$\|G_{\tau_k} * f\|_{L^r}^p \leq 2^p \|G_{\tau_0} * f\|_{L^r}^p + 2^p \left(\sum_{\ell=1}^k \|(G_{\tau_\ell} - G_{\tau_{\ell-1}}) * f\|_{L^r} \right)^p, \quad (18)$$

it follows that

$$\begin{aligned}
\sum_{k=0}^{\infty} \tau_k^{p\alpha/2} \|G_{\tau_k} * f\|_{L^r}^p &\leq 2^p \sum_{k=0}^{\infty} \tau_k^{p\alpha/2} \|G_{\tau_0} * f\|_{L^r}^p + 2^p \sum_{k=0}^{\infty} \left(\sum_{\ell=1}^k \tau_k^{\alpha/2} \tau_\ell^{-\alpha/2} \tau_\ell^{\alpha/2} \|(G_{\tau_\ell} - G_{\tau_{\ell-1}}) * f\|_{L^r} \right)^p \\
&\leq 2^p \frac{\tau_0^{p\alpha/2}}{1 - \omega^{p\alpha/2}} \|G_{\tau_0} * f\|_{L^r}^p + 2^p \left(\frac{1}{1 - \omega^{\alpha/2}} \right)^p \cdot \left(\sum_{\ell=1}^{\infty} \tau_\ell^{p\alpha/2} \|(G_{\tau_\ell} - G_{\tau_{\ell-1}}) * f\|_{L^r}^p \right), \quad (19)
\end{aligned}$$

and so

$$\begin{aligned}
\|f\|_{\Gamma_{p,r}^\alpha} &= \left(\sum_{k=0}^{\infty} \tau_k^{p\alpha/2} \|G_{\tau_k} * f\|_{L^r}^p \right)^{1/p} \\
&\leq 2 \left(\frac{1}{1 - \omega^{p\alpha/2}} \right)^{1/p} \tau_0^{\alpha/2} \|G_{\tau_0} * f\|_{L^r} + \left(\frac{2}{1 - \omega^{\alpha/2}} \right) \cdot \left(\sum_{\ell=1}^{\infty} \tau_\ell^{p\alpha/2} \|(G_{\tau_\ell} - G_{\tau_{\ell-1}}) * f\|_{L^r}^p \right)^{1/p} \\
&\leq \left(\frac{2}{1 - \omega^{\alpha/2}} \right) \tau_0^{\alpha/2} \|G_{\tau_0} * f\|_{L^r} + \left(\frac{2}{1 - \omega^{\alpha/2}} \right) \cdot \left(\sum_{\ell=1}^{\infty} \tau_\ell^{p\alpha/2} \|(G_{\tau_\ell} - G_{\tau_{\ell-1}}) * f\|_{L^r}^p \right)^{1/p} \\
&\leq \frac{C}{1 - \omega^{\alpha/2}} \cdot \|f\|_{\Gamma_{p,r}^\alpha}^{(1)}, \quad (20)
\end{aligned}$$

as claimed. $\square$

2.2 Choice of ω

The norms $\|f\|_{\Gamma_{p,r}^\alpha}$, and corresponding distances $D_{\alpha,p,r}(f, g)$, depend on a choice of rate parameter $0 < \omega < 1$ (which we suppress from the notation). It is therefore natural to ask about the relationships between these norms/distances for different choices of ω . It is easy to see that all choices of $0 < \omega < 1$ yield equivalent norms, in the sense that their ratios are bounded above and below by values independent of f ; such a proof can be extracted from any standard reference on Besov spaces [41, 46]. Indeed, the argument follows easily by showing that for any choice of ω , the norm $\|f\|_{\Gamma_{p,r}^\alpha}$ is equivalent to the continuous-scale norm

$$\left(\int_0^{\tau_0} t^{p\alpha/2-1} \|G_t * f\|_{L^r}^p dt \right)^{1/p}. \quad (21)$$

This may be shown by breaking this integral into integrals over the subintervals $[\tau_{k+1}, \tau_k]$ and obtaining a suitable upper or lower bound on each subinterval using monotonicity of the functions $t^{p\alpha/2-1}$ and $\|G_t * f\|_{L^r}^p$ (viewed as functions of t). We omit the straightforward details.

3 Key definitions

3.1 Deformations and maximum displacement

Suppose A and B are open subsets of $\mathbb{R}^d$, and $\varphi : B \rightarrow A$ is a C^1 bijection. If $f : A \rightarrow \mathbb{R}$ is in L^1 , then φ induces a new function $f_\varphi : B \rightarrow \mathbb{R}$ which we define as

$$f_\varphi(x) = f(\varphi(x))|\nabla\varphi(x)|, \quad (22)$$

where $\nabla\varphi(x)$ is the Jacobian matrix of φ , and $|\nabla\varphi(x)|$ denotes the absolute value of its determinant. We will call f_φ a deformation of f (and sometimes by abuse of terminology we may also refer to φ itself as a deformation). Note that f and f_φ have the same integrals and L^1 norms. Note too that if $\psi = \varphi^{-1}$ is the functional inverse of φ , then $f = (f_\varphi)_\psi$.

We define the maximum displacement of φ by

$$\varepsilon(\varphi) = \max_{x \in B} |x - \varphi(x)|. \quad (23)$$

Note too that $\varepsilon(\varphi) = \varepsilon(\varphi^{-1})$. This definition of maximum displacement is used in, for instance, the paper [22], and will be how we measure the “size” of the deformation φ .

Remark 4. The deformation f_φ preserves the L^1 norm and integral of f . This is natural in our setting, due to the relationship between the metrics $D_{\alpha,p,r}$ and Wasserstein distances (which are defined between probability measures, which obviously have equal measure). We do note, however, that other works have considered deformations of the form $f(\varphi(x))$, which instead preserve the L^∞ norm of f ; see [29, 1].

3.2 Mixed norms and projections

For a subspace $\mathcal{U} \leq \mathbb{R}^D$ of dimension d , we define the projection $\mathcal{P}_\mathcal{U}$ onto $\mathcal{U}$ by

$$(\mathcal{P}_\mathcal{U}f)(x) = \int_{\mathcal{U}^\perp} f(x, y) dy. \quad (24)$$

Projections of this type are fundamental in tomography [19, 5, 44, 47, 11, 51].

For a subspace $\mathcal{U}$ of $\mathbb{R}^D$ of dimension $d < D$, we define the mixed norm

$$\|f\|_{M^r(\mathcal{U})} = \left(\int_{\mathcal{U}} \left(\int_{\mathcal{U}^\perp} |f(x, y)| dy \right)^r dx \right)^{1/r}. \quad (25)$$

Note that

$$\|f\|_{M^r(\mathcal{U})} = \|\mathcal{P}_\mathcal{U}(|f|)\|_{L^r(\mathcal{U})}. \quad (26)$$

We also define the max-mixed norm

$$\|f\|_{M_d^r} = \sup_{\dim(\mathcal{U})=d} \|f\|_{M^r(\mathcal{U})}. \quad (27)$$

When d is clear from the context, we will just denote this by $\|f\|_{M^r}$.

Remark 5. Note that $\|f\|_{M^r}$ is rotationally-invariant, and that if f is compactly-supported, $\|f\|_{M^r}$ is controlled by the L^r norm of f . Indeed, if f is continuous and supported on, say, the ball $B_D(0, R)$ of radius R in $\mathbb{R}^D$, then each $f(x, \cdot)$ is supported on the ball $B_{D-d}(0, R)$ of radius R in $\mathcal{U}^\perp$ (identified with $\mathbb{R}^{D-d}$), and denoting its volume by $V_{D-d}(R)$,

$$\mathcal{P}_\mathcal{U}(|f|)(x) = \int_{\mathcal{U}^\perp} |f(x, y)| dy \leq V_{D-d}(R)^{1-1/r} \left(\int_{\mathcal{U}^\perp} |f(x, y)|^r dy \right)^{1/r}. \quad (28)$$

It then follows that

$$\|f\|_{M^r(\mathcal{U})} = \|\mathcal{P}_\mathcal{U}(|f|)\|_{L^r(\mathcal{U})} \leq V_{D-d}(R)^{1-1/r} \left(\int_{\mathcal{U}} \int_{\mathcal{U}^\perp} |f(x, y)|^r dy dx \right)^{1/r} = V_{D-d}(R)^{1-1/r} \|f\|_{L^r}, \quad (29)$$

and taking the supremum over $\mathcal{U}$ shows $\|f\|_{M^r} \leq V_{D-d}(R)^{1-1/r} \|f\|_{L^r}$ as well.

4 Properties of $D_{\alpha,p,r}$

4.1 Stability under deformations

In this section, we establish bounds on the distances $D_{\alpha,p,r}(f, f_\varphi)$ between a function f in L^r and its deformation $f_\varphi(x) = f(\varphi)|\nabla\varphi(x)|$.

The simplest bound is the following:

Proposition 4.1. *For any parameters $\alpha, p, r, \omega, \tau_0$, and any functions f and g in L^r ,*

$$D_{\alpha,p,r}(f, g) \leq \frac{\tau_0^{\alpha/2}}{1 - \omega^{\alpha/2}} \cdot \|f - g\|_{L^r}. \quad (30)$$

Proof. Using Young's inequality for convolutions (e.g. Theorem 8.7 in [15]) and $\|G_\tau\|_{L^1} = 1$,

$$D_{\alpha,p,r}(f, g)^p = \sum_{k \geq 0} \tau_k^{p\alpha/2} \|G_{\tau_k} * (f - g)\|_{L^r}^p \leq \|f - g\|_{L^r}^p \sum_{k \geq 0} \tau_k^{p\alpha/2} = \frac{\tau_0^{p\alpha/2}}{1 - \omega^{p\alpha/2}} \|f - g\|_{L^r}^p, \quad (31)$$

and the result follows by taking p -th roots and using Lemma 2.2. $\square$

Consequently, the metric $D_{\alpha,p,r}(f, g)$ is no less stable than $\|f - g\|_{L^r}$. However, we now state and prove a bound that controls the distance $D_{\alpha,p,r}(f, f_\varphi)$ by the deformation size $\varepsilon(\varphi)$. Similar bounds are known (and are essentially tautological) for Wasserstein distances, and are also known for other distances such as the Cramér and sliced Cramér metrics [22].

For $\alpha > 0$ and $p \geq 1$, define the function $\Delta_{\alpha,p}(\epsilon)$, $0 < \epsilon < 1$, as follows:

$$\Delta_{\alpha,p}(\epsilon) = \begin{cases} \frac{\epsilon^\alpha}{(\alpha - \alpha^2)^{1/p}}, & \text{if } 0 < \alpha < 1, \\ \epsilon \cdot (\log(1/\epsilon) + 1)^{1/p}, & \text{if } \alpha = 1, \\ \frac{\epsilon}{\min\{(\alpha - 1)^{1/p}, 1\}}, & \text{if } \alpha > 1. \end{cases} \quad (32)$$

Theorem 4.2. *Let $A, B \subset \mathbb{R}^D$ be open sets, and let $\varphi : B \rightarrow A$ be a C^1 bijection. Let $d \leq D$ and $\mathcal{U}$ be a d -dimensional subspace of $\mathbb{R}^D$. Suppose f is in $L^1 \cap L^r$ and supported on A , with $\|f_\varphi\|_{M^r(\mathcal{U})} \leq \|f\|_{M^r(\mathcal{U})} < \infty$, where $r \geq 1$. Let $\varepsilon = \varepsilon(\varphi) = \max_{x \in B} |x - \varphi(x)|$, and suppose $\varepsilon < \sqrt{\tau_0}$. Then for all $\alpha > 0$ and $p \geq 1$,*

$$D_{\alpha,p,r}(\mathcal{P}_\mathcal{U} f, \mathcal{P}_\mathcal{U} f_\varphi) \leq C \cdot \|f\|_{M^r(\mathcal{U})} \cdot \frac{\tau_0^{\alpha/2}}{1 - \omega^{1/2}} \cdot \Delta_{\alpha,p}(\varepsilon/\sqrt{\tau_0}) \quad (33)$$

where $C > 0$ depends only on the dimension d .

As an immediate corollary without projections, we have:

Corollary 4.3. *Let $A, B \subset \mathbb{R}^d$ be open sets, and let $\varphi : B \rightarrow A$ be a C^1 bijection. Suppose f is in L^r and supported on A , and satisfies $\|f_\varphi\|_{L^r} \leq \|f\|_{L^r} < \infty$, where $r \geq 1$. Let $\varepsilon = \varepsilon(\varphi) = \max_{x \in B} |x - \varphi(x)|$, and suppose $\varepsilon < \sqrt{\tau_0}$. Then for all $\alpha > 0$ and $p \geq 1$,*

$$D_{\alpha,p,r}(f, f_\varphi) \leq C \cdot \|f\|_{L^r} \cdot \frac{\tau_0^{\alpha/2}}{1 - \omega^{1/2}} \cdot \Delta_{\alpha,p}(\varepsilon/\sqrt{\tau_0}) \quad (34)$$

where $C > 0$ depends only on the dimension d .

The proof of Theorem 4.2 follows from the following bounds on the distances at each sufficiently coarse level:

Proposition 4.4. *Adopt the notation of Theorem 4.2, let $\tau > 0$, and suppose $\varepsilon = \varepsilon(\varphi) \leq \sqrt{\tau}$. Then*

$$\|G_\tau * (\mathcal{P}_\mathcal{U} f - \mathcal{P}_\mathcal{U} f_\varphi)\|_{L^r} \leq C \cdot \|f\|_{M^r(\mathcal{U})} \cdot \frac{\varepsilon}{\sqrt{\tau}}, \quad (35)$$

where $C > 0$ depends on the dimension d .

We will make use of the following lemma:

Lemma 4.5. *Suppose $0 < \delta \leq 1$, and $|\xi(y)| \leq \delta$ for all y in $\mathbb{R}^d$. Let $G = G_1$. Then*

$$\int_{\mathbb{R}^d} |G(y) - G(y + \xi(y))| dy \leq 2 \frac{\Gamma((d+1)/2)}{\Gamma(d/2)} \delta + O(\delta^2). \quad (36)$$

Proof of Lemma 4.5. We have the factorization

$$\begin{aligned}
G(y + \xi(y)) &= \frac{1}{\pi^{d/2}} e^{-|y + \xi(y)|^2} \\
&= \frac{1}{\pi^{d/2}} e^{-|y|^2} e^{-(2\langle y, \xi(y) \rangle + |\xi(y)|^2)} \\
&= G(y) e^{-|\xi(y)|^2} e^{-2\langle y, \xi(y) \rangle},
\end{aligned} \tag{37}$$

and so

$$\begin{aligned}
\int |G(y) - G(y + \xi(y))| dy &= \int \left| G(y) e^{-|\xi(y)|^2} e^{|\xi(y)|^2} - G(y) e^{-|\xi(y)|^2} e^{-2\langle y, \xi(y) \rangle} \right| dy \\
&= \int G(y) e^{-|\xi(y)|^2} \left| e^{|\xi(y)|^2} - e^{-2\langle y, \xi(y) \rangle} \right| dy \\
&\leq \int G(y) \left| e^{|\xi(y)|^2} - 1 + 1 - e^{-2\langle y, \xi(y) \rangle} \right| dy \\
&= \int G(y) \left| e^{|\xi(y)|^2} - 1 + 1 - e^{-2\langle y, \xi(y) \rangle} \right| dy \\
&\leq \int G(y) \left| e^{|\xi(y)|^2} - 1 \right| dy + \int G(y) \left| e^{-2\langle y, \xi(y) \rangle} - 1 \right| dy.
\end{aligned} \tag{38}$$

For the first integral, since $|\xi(y)| \leq \delta \leq 1$, there is a universal constant $C > 0$ such that for all y ,

$$e^{|\xi(y)|^2} - 1 = C|\xi(y)|^2 \leq C\delta^2, \tag{39}$$

and so, since $\int G(y) dy = 1$,

$$\int G(y) \left| e^{|\xi(y)|^2} - 1 \right| dy \leq C\delta^2. \tag{40}$$

For the second integral: we have the Taylor expansion

$$e^{-2\langle y, \xi(y) \rangle} - 1 = \sum_{k=1}^{\infty} \frac{(-2\langle y, \xi(y) \rangle)^k}{k!}, \tag{41}$$

and so, since $|\langle y, \xi(y) \rangle| \leq |y||\xi(y)| \leq \delta|y|$,

$$\left| e^{-2\langle y, \xi(y) \rangle} - 1 \right| \leq \sum_{k=1}^{\infty} \frac{(2\delta)^k}{k!} |y|^k. \tag{42}$$

Now, for each $k \geq 1$, changing into spherical coordinates and using that the area of the sphere of radius r in $\mathbb{R}^d$ is

$$A_{d-1}(r) = r^{d-1} \frac{2\pi^{d/2}}{\Gamma(d/2)}, \tag{43}$$

we get

$$\begin{aligned}
\int G(y) |y|^k dy &= \int \frac{e^{-|y|^2}}{\pi^{d/2}} |y|^k dy \\
&= \frac{2\pi^{d/2}}{\Gamma(d/2)} \int_0^{\infty} r^{d-1} r^k \frac{e^{-r^2}}{\pi^{d/2}} dr \\
&= \frac{2}{\Gamma(d/2)} \int_0^{\infty} r^{d+k-1} e^{-r^2} dr, \\
&= \frac{2}{\Gamma(d/2)} \int_0^{\infty} r^{d+k-2} e^{-r^2} r dr,
\end{aligned} \tag{44}$$

and using the change-of-variables $s = r^2$, $ds = 2r dr$,

$$\begin{aligned}
\int G(y) |y|^k dy &= \frac{2}{\Gamma(d/2)} \int_0^{\infty} r^{d+k-2} e^{-r^2} r dr \\
&= \frac{1}{\Gamma(d/2)} \int_0^{\infty} s^{(d+k)/2-1} e^{-s} ds \\
&= \frac{\Gamma((d+k)/2)}{\Gamma(d/2)}.
\end{aligned} \tag{45}$$

Consequently, we have the Taylor series

$$\begin{aligned}
\int G(y) \left| e^{-2\langle y, \xi(y) \rangle} - 1 \right| dy &\leq \sum_{k=1}^{\infty} \frac{(2\delta)^k}{k!} \int G(y) |y|^k dy \\
&= \sum_{k=1}^{\infty} \frac{\Gamma((d+k)/2)}{\Gamma(d/2)k!} (2\delta)^k \\
&= 2 \frac{\Gamma((d+1)/2)}{\Gamma(d/2)} \delta + O(\delta^2).
\end{aligned} \tag{46}$$

Combining (46) with (39) gives the desired bound

$$\int_{\mathbb{R}^d} |G(y) - G(y + \xi(y))| dy \leq 2 \frac{\Gamma((d+1)/2)}{\Gamma(d/2)} \delta + O(\delta^2). \tag{47}$$

□

Proof of Proposition 4.4. We recall that we wish to show the bound

$$\|G_\tau * (\mathcal{P}_U f - \mathcal{P}_U f_\varphi)\|_{L^r} \leq C \cdot \|f\|_{M^r(U)} \cdot \frac{\varepsilon}{\sqrt{\tau}}, \tag{48}$$

for $\varepsilon \leq \sqrt{\tau}$. Without loss of generality, we identify the d -dimensional subspace $\mathcal{U}$ with $\mathbb{R}^d$; and for brevity, we let $\mathcal{P} = \mathcal{P}_U$. First, letting $\psi = \varphi^{-1}$, and $\psi(y) = (\psi_1, \psi_2) \in \mathbb{R}^d \times \mathbb{R}^{D-d}$,

$$\begin{aligned}
(G_\tau * \mathcal{P} f_\varphi)(x) &= \int_{\mathbb{R}^d} G_\tau(x-w) (\mathcal{P} f_\varphi)(w) dw \\
&= \int_{\mathbb{R}^d} G_\tau(x-w) \int_{\mathbb{R}^{D-d}} f_\varphi(w, y) dy dw \\
&= \int_{\mathbb{R}^d} \int_{\mathbb{R}^{D-d}} G_\tau(x-w) f(\varphi(w, y)) |\nabla \varphi(w, y)| dy dw \\
&= \int_{\mathbb{R}^d} \int_{\mathbb{R}^{D-d}} G_\tau(x - \psi_1(u, v)) f(u, v) dv du
\end{aligned} \tag{49}$$

and similarly

$$(G_\tau * \mathcal{P} f)(x) = \int_{\mathbb{R}^d} \int_{\mathbb{R}^{D-d}} G_\tau(x-u) f(u, v) dv du. \tag{50}$$

Therefore,

$$(G_\tau * [\mathcal{P} f - \mathcal{P} f_\varphi])(x) = \int_{\mathbb{R}^d} \int_{\mathbb{R}^{D-d}} [G_\tau(x-u) - G_\tau(x - \psi_1(u, v))] f(u, v) dv du, \tag{51}$$

and so

$$\begin{aligned}
|(G_\tau * [\mathcal{P} f - \mathcal{P} f_\varphi])(x)| &\leq \int_{\mathbb{R}^d} \int_{\mathbb{R}^{D-d}} |G_\tau(x-u) - G_\tau(x - \psi_1(u, v))| \cdot |f(u, v)| dv du, \\
&\leq \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^{D-d}} |f(u, v)| dv \right) \left(\sup_{v \in \mathbb{R}^{D-d}} |G_\tau(x-u) - G_\tau(x - \psi_1(u, v))| \right) du \\
&= \int_{\mathbb{R}^d} (\mathcal{P}|f|)(u) \left(\sup_{v \in \mathbb{R}^{D-d}} |G_\tau(x-u) - G_\tau(x - \psi_1(u, v))| \right) du.
\end{aligned} \tag{52}$$

Since $|\psi_1(u, v) - u| \leq \varepsilon$,

$$\sup_{v \in \mathbb{R}^{D-d}} |G_\tau(x-u) - G_\tau(x - \psi_1(u, v))| \leq \max_{w \in B(u, \varepsilon)} |G_\tau(x-u) - G_\tau(x-w)|, \tag{53}$$

noting that the maximum is achieved since the function being maximized is continuous and the closed ball $B(u, \varepsilon)$ is compact; we therefore define

$$w^*(x, u) = \operatorname{argmax}_{w \in B(u, \varepsilon)} |G_\tau(x-u) - G_\tau(x-w)|. \tag{54}$$

For brevity, let

$$K_\tau(x, u) = |G_\tau(x-u) - G_\tau(x-w^*(x, u))|. \tag{55}$$

We then wish to bound the L^r norm of the function

$$x \mapsto \int_{\mathbb{R}^d} K_\tau(x, u) h(u) du, \quad (56)$$

where $h(u) = \mathcal{P}|f|$. We will follow a standard argument, by first bounding the integrals $\int K_\tau(x, u) du$ and $\int K_\tau(x, u) dx$, making use of Lemma 4.5.

After a change of variables $y = (x - u)/\sqrt{\tau}$, i.e. $x = \sqrt{\tau}y + u$,

$$\begin{aligned} \int K_\tau(x, u) dx &= \int |G_\tau(x - u) - G_\tau(x - w^*(x, u))| dx = \int \left| G(y) - G\left(\frac{\sqrt{\tau}y + u - w^*(\sqrt{\tau}y + u, u)}{\sqrt{\tau}}\right) \right| dy \\ &= \int \left| G(y) - G\left(y + \frac{u - w^*(\sqrt{\tau}y + u, u)}{\sqrt{\tau}}\right) \right| dy \\ &= \int |G(y) - G(y + \xi(y))| dy, \end{aligned} \quad (57)$$

where

$$\xi(y) = \frac{u - w^*(\sqrt{\tau}y + u, u)}{\sqrt{\tau}}. \quad (58)$$

Note that, since $w^*(x, u) \in B(u, \varepsilon)$,

$$|\xi(y)| \leq \frac{\varepsilon}{\sqrt{\tau}}. \quad (59)$$

From Lemma 4.5, therefore,

$$\int |G_\tau(x - u) - G_\tau(x - w^*(x, u))| dx \leq 2 \frac{\Gamma((d+1)/2)}{\Gamma(d/2)} \frac{\varepsilon}{\sqrt{\tau}} + O\left(\left(\frac{\varepsilon}{\sqrt{\tau}}\right)^2\right). \quad (60)$$

Next, we bound the other integral, using a nearly identical argument. Indeed, after a change of variables $y = (x - u)/\sqrt{\tau}$, i.e. $u = x - \sqrt{\tau}y$,

$$\begin{aligned} \int K_\tau(x, u) du &= \int |G_\tau(x - u) - G_\tau(x - w^*(x, u))| du = \int \left| G(y) - G\left(\frac{x - w^*(x, x - \sqrt{\tau}y)}{\sqrt{\tau}}\right) \right| dy \\ &= \int \left| G(y) - G\left(y + \frac{x - \sqrt{\tau}y - w^*(x, x - \sqrt{\tau}y)}{\sqrt{\tau}}\right) \right| dy \\ &= \int |G(y) - G(y + \xi(y))| dy, \end{aligned} \quad (61)$$

where

$$\xi(y) = \frac{x - \sqrt{\tau}y - w^*(x, x - \sqrt{\tau}y)}{\sqrt{\tau}}. \quad (62)$$

Note that, since $w^*(x, x - \sqrt{\tau}y) \in B(x - \sqrt{\tau}y, \varepsilon)$,

$$|\xi(y)| \leq \frac{\varepsilon}{\sqrt{\tau}}. \quad (63)$$

From Lemma 4.5, therefore,

$$\int |G_\tau(x - u) - G_\tau(x - w^*(x, u))| du \leq 2 \frac{\Gamma((d+1)/2)}{\Gamma(d/2)} \frac{\varepsilon}{\sqrt{\tau}} + O\left(\left(\frac{\varepsilon}{\sqrt{\tau}}\right)^2\right). \quad (64)$$

It then follows (see, e.g. Theorem 6.18 in [15]) that the L^r operator norm of the kernel K_τ is bounded above

$$\|K_\tau\|_{L^r \rightarrow L^r} \leq 2 \frac{\Gamma((d+1)/2)}{\Gamma(d/2)} \frac{\varepsilon}{\sqrt{\tau}} + O\left(\left(\frac{\varepsilon}{\sqrt{\tau}}\right)^2\right), \quad (65)$$

and as a consequence,

$$\|G_\tau * [\mathcal{P}f - \mathcal{P}_\varphi]\|_{L^r} \leq \left(\left| \int (\mathcal{P}|f|)(u) K_\tau(x, u) du \right|^r dx \right)^{1/r} \leq C \|\mathcal{P}(|f|)\|_{L^r} \frac{\varepsilon}{\sqrt{\tau}} = C \cdot \|f\|_{M^r(\mathcal{U})} \cdot \frac{\varepsilon}{\sqrt{\tau}}, \quad (66)$$

as desired. $\square$

We now employ Proposition 4.4 to prove Theorem 4.2. We first state an elementary lemma, which we prove for the reader's convenience.

Lemma 4.6. *For all $0 < \eta < 1$ and for all $0 < \beta < 1$,*

$$\frac{1}{1-\eta^\beta} \leq \frac{1}{1-\eta} \cdot \frac{1}{\beta}. \quad (67)$$

Proof of Lemma 4.6. The desired inequality is equivalent to

$$\beta\eta - \eta^\beta + 1 - \beta \geq 0. \quad (68)$$

Fix $0 < \beta < 1$, and denote the function on the left by $H(\eta)$. Then $H(0) = 1 - \beta > 0$, and $H(1) = \beta - 1 + 1 - \beta = 0$. Furthermore,

$$H'(\eta) = \beta - \beta\eta^{\beta-1} = \beta(1 - \eta^{\beta-1}) < 0, \quad (69)$$

since $0 < \eta < 1$ and $\beta - 1 < 0$; therefore, $H(\eta)$ decreases from $H(0) = 1 - \beta$ to $H(1) = 0$, and hence must always be positive, as desired. $\square$

Proof of Theorem 4.2. Let

$$M = \lfloor \log_{1/\omega}(\tau_0/\varepsilon^2) \rfloor. \quad (70)$$

We break the proof into the three regimes of α appearing in the definition of $\Delta_{\alpha,p}$.

Case 1: $0 < \alpha < 1$. If $k \leq M$, then $\varepsilon/\sqrt{\tau_k} \leq 1$, and so, by Proposition 4.4,

$$\|G_{\tau_k} * [\mathcal{P}f - \mathcal{P}f_\varphi]\|_{L^r} \leq C \cdot \|f\|_{M^r(\mathcal{U})} \cdot \frac{\varepsilon}{\sqrt{\tau_k}}, \quad (71)$$

where C depends only on the dimension d . Therefore,

$$\begin{aligned} \sum_{k=0}^M \tau_k^{p\alpha/2} \|G_{\tau_k} * [\mathcal{P}f - \mathcal{P}f_\varphi]\|_{L^r}^p &\leq C^p \cdot \|f\|_{M^r(\mathcal{U})}^p \cdot \sum_{k=0}^M \tau_k^{p\alpha/2} \frac{\varepsilon^p}{\tau_k^{p/2}} \\ &\leq C^p \cdot \|f\|_{M^r(\mathcal{U})}^p \cdot \varepsilon^p \cdot \tau_0^{(\alpha-1)p/2} \sum_{k=0}^M \omega^{k(\alpha-1)p/2} \\ &\leq C^p \cdot \|f\|_{M^r(\mathcal{U})}^p \cdot \varepsilon^p \cdot \tau_0^{(\alpha-1)p/2} \frac{\omega^{M(\alpha-1)p/2}}{1 - \omega^{(1-\alpha)p/2}} \\ &\leq C^p \cdot \|f\|_{M^r(\mathcal{U})}^p \cdot \varepsilon^p \cdot \frac{\varepsilon^{(\alpha-1)p}}{1 - \omega^{(1-\alpha)p/2}} \\ &= C^p \cdot \|f\|_{M^r(\mathcal{U})}^p \cdot \varepsilon^{p\alpha} \cdot \frac{1}{1 - \omega^{(1-\alpha)p/2}}, \end{aligned} \quad (72)$$

and so using Lemma 4.6,

$$\sum_{k=0}^M \tau_k^{p\alpha/2} \|G_{\tau_k} * [\mathcal{P}f - \mathcal{P}f_\varphi]\|_{L^r}^p \leq C^p \cdot \|f\|_{M^r(\mathcal{U})}^p \cdot \varepsilon^{p\alpha} \cdot \frac{1}{1 - \omega^{p/2}} \cdot \frac{1}{1 - \alpha}. \quad (73)$$

For the remainder of the series, we use the simpler bound $\|G_{\tau_k} * [\mathcal{P}f - \mathcal{P}f_\varphi]\|_{L^r} \leq \|\mathcal{P}f - \mathcal{P}f_\varphi\|_{L^r} \leq 2\|f\|_{M^r(\mathcal{U})}$, and get

$$\begin{aligned} \sum_{k=M+1}^{\infty} \tau_k^{p\alpha/2} \|G_{\tau_k} * [\mathcal{P}f - \mathcal{P}f_\varphi]\|_{L^r}^p &\leq 2^p \|f\|_{M^r(\mathcal{U})}^p \sum_{k=M+1}^{\infty} \tau_k^{p\alpha/2} \\ &\leq 2^p \|f\|_{M^r(\mathcal{U})}^p \tau_0^{p\alpha/2} \sum_{k=M+1}^{\infty} \omega^{kp\alpha/2} \\ &= 2^p \|f\|_{M^r(\mathcal{U})}^p \tau_0^{p\alpha/2} \frac{\omega^{(M+1)p\alpha/2}}{1 - \omega^{p\alpha/2}} \\ &\leq 2^p \|f\|_{M^r(\mathcal{U})}^p \cdot \frac{1}{1 - \omega^{p\alpha/2}} \cdot \varepsilon^{p\alpha} \end{aligned} \quad (74)$$

and again using Lemma 4.6,

$$\sum_{k=M+1}^{\infty} \tau_k^{p\alpha/2} \|G_{\tau_k} * [\mathcal{P}f - \mathcal{P}f_\varphi]\|_{L^r}^p \leq 2^p \|f\|_{M^r(\mathcal{U})}^p \cdot \varepsilon^{p\alpha} \cdot \frac{1}{1 - \omega^{p/2}} \cdot \frac{1}{\alpha}. \quad (75)$$

Adding the bounds (73) and (75) gives

$$\begin{aligned}
D_{\alpha,p,r}(f, f_\varphi)^p &\leq C^p \cdot \|f\|_{M^r(\mathcal{U})}^p \cdot \varepsilon^{p\alpha} \cdot \frac{1}{1-\omega^{p/2}} \cdot \frac{1}{1-\alpha} + 2^p \|f\|_{M^r(\mathcal{U})}^p \cdot \varepsilon^{p\alpha} \cdot \frac{1}{1-\omega^{p/2}} \cdot \frac{1}{\alpha} \\
&= C^p \cdot \|f\|_{M^r(\mathcal{U})}^p \cdot \frac{1}{1-\omega^{p/2}} \cdot \left(\frac{1}{1-\alpha} + \frac{1}{\alpha} \right) \cdot \varepsilon^{p\alpha} \\
&= C^p \cdot \|f\|_{M^r(\mathcal{U})}^p \cdot \frac{1}{1-\omega^{p/2}} \cdot \frac{1}{\alpha(1-\alpha)} \cdot \varepsilon^{p\alpha},
\end{aligned} \tag{76}$$

and taking a p -th root and using Lemma 2.2 gives

$$\begin{aligned}
D_{\alpha,p,r}(f, f_\varphi) &\leq C \cdot \|f\|_{M^r(\mathcal{U})} \cdot \frac{1}{1-\omega^{1/2}} \cdot \frac{1}{(\alpha-\alpha^2)^{1/p}} \cdot \varepsilon^\alpha \\
&= C \cdot \|f\|_{M^r(\mathcal{U})} \cdot \frac{\tau_0^{\alpha/2}}{1-\omega^{1/2}} \cdot \frac{1}{(\alpha-\alpha^2)^{1/p}} \cdot \varepsilon^\alpha / \tau_0^{\alpha/2} \\
&= C \cdot \|f\|_{M^r(\mathcal{U})} \cdot \frac{\tau_0^{\alpha/2}}{1-\omega^{1/2}} \cdot \Delta_{\alpha,p}(\varepsilon/\sqrt{\tau_0}).
\end{aligned} \tag{77}$$

Case 2: $\alpha = 1$. If $k \leq M$, then $\varepsilon/\sqrt{\tau_k} \leq 1$, and so, by Proposition 4.4,

$$\|G_{\tau_k} * [\mathcal{P}f - \mathcal{P}f_\varphi]\|_{L^r} \leq C \cdot \|f\|_{M^r(\mathcal{U})} \cdot \frac{\varepsilon}{\sqrt{\tau_k}}, \tag{78}$$

where C depends only on the dimension d . Therefore,

$$\begin{aligned}
\sum_{k=0}^M \tau_k^{p/2} \|G_{\tau_k} * [\mathcal{P}f - \mathcal{P}f_\varphi]\|_{L^r}^p &\leq C^p \cdot \|f\|_{M^r(\mathcal{U})}^p \cdot \sum_{k=0}^M \tau_k^{p/2} \frac{\varepsilon^p}{\tau_k^{p/2}} \\
&\leq C^p \cdot \|f\|_{M^r(\mathcal{U})}^p \cdot \varepsilon^p \cdot M \\
&\leq C^p \cdot \|f\|_{M^r(\mathcal{U})}^p \cdot \varepsilon^p \cdot \log_{1/\omega}(\tau_0/\varepsilon^2).
\end{aligned} \tag{79}$$

For the remainder of the series, we use the elementary bound

$$\log(1/t) \geq 1-t, \quad 0 < t \leq 1. \tag{80}$$

Indeed, the two sides agree at $t = 1$ (where they are both 0); and the difference $\log(1/t) - 1 + t$ has derivative $-1/t + 1 \leq 0$.

Turning to the series, we again use the simpler bound $\|G_{\tau_k} * [\mathcal{P}f - \mathcal{P}f_\varphi]\|_{L^r} \leq 2\|f\|_{M^r(\mathcal{U})}$, and get

$$\begin{aligned}
\sum_{k=M+1}^{\infty} \tau_k^{p/2} \|G_{\tau_k} * [\mathcal{P}f - \mathcal{P}f_\varphi]\|_{L^r}^p &\leq 2^p \|f\|_{M^r(\mathcal{U})}^p \tau_0^{p/2} \sum_{k=M+1}^{\infty} \omega^{kp/2} \\
&= 2^p \|f\|_{M^r(\mathcal{U})}^p \tau_0^{p/2} \frac{\omega^{(M+1)p/2}}{1-\omega^{p/2}} \\
&\leq 2^p \|f\|_{M^r(\mathcal{U})}^p \cdot \frac{1}{1-\omega^{p/2}} \cdot \varepsilon^p.
\end{aligned} \tag{81}$$

Adding the bounds (79) and (81), and using that $(p/2)\log(1/\omega) = \log(1/\omega^{p/2}) \geq 1 - \omega^{p/2}$, gives

$$\begin{aligned}
D_{1,p,r}(f, f_\varphi)^p &\leq C^p \cdot \|f\|_{M^r(\mathcal{U})}^p \cdot \varepsilon^p \cdot \log_{1/\omega}(\tau_0/\varepsilon^2) + 2^p \|f\|_{M^r(\mathcal{U})}^p \cdot \frac{1}{1-\omega^{p/2}} \cdot \varepsilon^p \\
&= C^p \cdot \|f\|_{M^r(\mathcal{U})}^p \cdot \frac{\varepsilon^p}{\tau_0^{p/2}} \cdot \log(\tau_0/\varepsilon^2) \cdot \frac{\tau_0^{p/2}}{\log(1/\omega)} + 2^p \|f\|_{M^r(\mathcal{U})}^p \cdot \frac{\tau_0^{p/2}}{1-\omega^{p/2}} \cdot \frac{\varepsilon^p}{\tau_0^{p/2}} \\
&= C^p \cdot \|f\|_{M^r(\mathcal{U})}^p \cdot \frac{\varepsilon^p}{\tau_0^{p/2}} \cdot \left(\log(\tau_0/\varepsilon^2) \cdot \frac{\tau_0^{p/2}}{\log(1/\omega)} + \frac{\tau_0^{p/2}}{1-\omega^{p/2}} \right) \\
&= C^p \cdot \|f\|_{M^r(\mathcal{U})}^p \cdot \frac{\varepsilon^p}{\tau_0^{p/2}} \cdot \left(\log(\tau_0/\varepsilon^2) \cdot \frac{(p/2)\tau_0^{p/2}}{(p/2)\log(1/\omega)} + \frac{\tau_0^{p/2}}{1-\omega^{p/2}} \right) \\
&= C^p \cdot \|f\|_{M^r(\mathcal{U})}^p \cdot \tau_0^p \cdot \frac{\varepsilon^p}{\tau_0^{p/2}} \cdot \left(\log(\sqrt{\tau_0}/\varepsilon) \cdot \frac{p}{1-\omega^{p/2}} + \frac{1}{1-\omega^{p/2}} \right) \\
&\leq C^p \cdot \|f\|_{M^r(\mathcal{U})}^p \cdot \tau_0^{p/2} \cdot \frac{\varepsilon^p}{\tau_0^{p/2}} \cdot (\log(\sqrt{\tau_0}/\varepsilon) + 1) \cdot \frac{p}{1-\omega^{p/2}},
\end{aligned} \tag{82}$$

and taking a p -th root and using Lemma 2.2 and that $p^{1/p}$ is bounded gives

$$\begin{aligned} D_{1,p,r}(f, f_\varphi)^p &\leq C \cdot \|f\|_{M^r(\mathcal{U})} \cdot \frac{\tau_0^{1/2}}{1 - \omega^{1/2}} \cdot \frac{\varepsilon}{\tau_0^{1/2}} \cdot (\log(\sqrt{\tau_0}/\varepsilon) + 1)^{1/p} \\ &= C \cdot \|f\|_{M^r(\mathcal{U})} \cdot \frac{\tau_0^{1/2}}{1 - \omega^{1/2}} \cdot \Delta_{1,p}(\varepsilon/\sqrt{\tau_0}). \end{aligned} \quad (83)$$

Case 3: $\alpha > 1$. If $k \leq M$, then $\varepsilon/\sqrt{\tau_k} \leq 1$, and so, by Proposition 4.4,

$$\|G_{\tau_k} * [\mathcal{P}f - \mathcal{P}f_\varphi]\|_{L^r} \leq C \cdot \|f\|_{M^r(\mathcal{U})} \cdot \frac{\varepsilon}{\sqrt{\tau_k}}, \quad (84)$$

where C only depends on the dimension d . Therefore,

$$\begin{aligned} \sum_{k=0}^M \tau_k^{p\alpha/2} \|G_{\tau_k} * [\mathcal{P}f - \mathcal{P}f_\varphi]\|_{L^r}^p &\leq C^p \cdot \|f\|_{M^r(\mathcal{U})}^p \cdot \sum_{k=0}^M \tau_k^{p\alpha/2} \cdot \frac{\varepsilon^p}{\tau_k^{p/2}} \\ &\leq C^p \cdot \|f\|_{M^r(\mathcal{U})}^p \cdot \varepsilon^p \cdot \tau_0^{(\alpha-1)p/2} \sum_{k=0}^M \omega^{k(\alpha-1)p/2} \\ &\leq C^p \cdot \|f\|_{M^r(\mathcal{U})}^p \cdot \varepsilon^p \cdot \frac{\tau_0^{(\alpha-1)p/2}}{1 - \omega^{(\alpha-1)p/2}}. \end{aligned} \quad (85)$$

For the remainder of the series, we again use the simpler bound $\|G_{\tau_k} * [\mathcal{P}f - \mathcal{P}f_\varphi]\|_{L^r} \leq 2\|f\|_{M^r(\mathcal{U})}$, and, using that $(\varepsilon^2/\tau_0)^\alpha \leq \varepsilon^2/\tau_0$, we get

$$\begin{aligned} \sum_{k=M+1}^{\infty} \tau_k^{p\alpha/2} \|G_{\tau_k} * [\mathcal{P}f - \mathcal{P}f_\varphi]\|_{L^r}^p &\leq 2^p \|f\|_{M^r(\mathcal{U})}^p \tau_0^{p\alpha/2} \sum_{k=M+1}^{\infty} \omega^{kp\alpha/2} \\ &= 2^p \|f\|_{M^r(\mathcal{U})}^p \tau_0^{p\alpha/2} \frac{\omega^{(M+1)p\alpha/2}}{1 - \omega^{p\alpha/2}} \\ &\leq 2^p \|f\|_{M^r(\mathcal{U})}^p \tau_0^{p\alpha/2} \frac{(\varepsilon^2/\tau_0)^{p\alpha/2}}{1 - \omega^{p\alpha/2}} \\ &\leq 2^p \|f\|_{M^r(\mathcal{U})}^p \tau_0^{p\alpha/2} \frac{(\varepsilon^2/\tau_0)^{p/2}}{1 - \omega^{p\alpha/2}} \\ &\leq 2^p \|f\|_{M^r(\mathcal{U})}^p \cdot \varepsilon^p \cdot \frac{\tau_0^{(\alpha-1)p/2}}{1 - \omega^{p\alpha/2}}. \end{aligned} \quad (86)$$

Adding the bounds (85) and (86),

$$\begin{aligned} D_{\alpha,p,r}(f, f_\varphi)^p &\leq C^p \cdot \|f\|_{M^r(\mathcal{U})}^p \cdot \varepsilon^p \cdot \frac{\tau_0^{(\alpha-1)p/2}}{1 - \omega^{(\alpha-1)p/2}} + 2^p \|f\|_{M^r(\mathcal{U})}^p \cdot \varepsilon^p \cdot \frac{\tau_0^{(\alpha-1)p/2}}{1 - \omega^{p\alpha/2}} \\ &= C^p \cdot \|f\|_{M^r(\mathcal{U})}^p \cdot \frac{\varepsilon^p}{\tau_0^{p/2}} \cdot \tau_0^{\alpha p/2} \cdot \left(\frac{1}{1 - \omega^{(\alpha-1)p/2}} + \frac{1}{1 - \omega^{p\alpha/2}} \right), \end{aligned} \quad (87)$$

and taking p -th roots,

$$\begin{aligned} D_{\alpha,p,r}(f, f_\varphi) &\leq C \cdot \|f\|_{M^r(\mathcal{U})} \cdot \frac{\varepsilon}{\tau_0^{1/2}} \cdot \tau_0^{\alpha/2} \cdot \left(\frac{1}{1 - \omega^{(\alpha-1)p/2}} + \frac{1}{1 - \omega^{p\alpha/2}} \right)^{1/p} \\ &\leq C \cdot \|f\|_{M^r(\mathcal{U})} \cdot \frac{\varepsilon}{\tau_0^{1/2}} \cdot \tau_0^{\alpha/2} \cdot \left[\left(\frac{1}{1 - \omega^{(\alpha-1)p/2}} \right)^{1/p} + \left(\frac{1}{1 - \omega^{p\alpha/2}} \right)^{1/p} \right] \\ &\leq C \cdot \|f\|_{M^r(\mathcal{U})} \cdot \frac{\varepsilon}{\tau_0^{1/2}} \cdot \tau_0^{\alpha/2} \cdot \left[\left(\frac{1}{1 - \omega^{(\alpha-1)p/2}} \right)^{1/p} + \frac{1}{1 - \omega^{1/2}} \right]. \end{aligned} \quad (88)$$

For the last term, consider two cases. When $1 < \alpha < 2$, we have the bound

$$\frac{1}{1 - \omega^{(\alpha-1)p/2}} \leq \frac{1}{1 - \omega^{p/2}} \cdot \frac{1}{\alpha - 1}, \quad (89)$$

and so

$$\left(\frac{1}{1 - \omega^{(\alpha-1)p/2}} \right)^{1/p} \leq \frac{1}{1 - \omega^{1/2}} \cdot \frac{1}{(\alpha - 1)^{1/p}}, \quad (90)$$

whereas for $\alpha \geq 2$, we have the simpler bound

$$\left(\frac{1}{1 - \omega^{(\alpha-1)p/2}}\right)^{1/p} \leq \left(\frac{1}{1 - \omega^{p/2}}\right)^{1/p} \leq \frac{1}{1 - \omega^{1/2}}. \quad (91)$$

Putting these cases together,

$$\left(\frac{1}{1 - \omega^{(\alpha-1)p/2}}\right)^{1/p} \leq \frac{1}{1 - \omega^{1/2}} \cdot \frac{1}{\min\{1, (\alpha-1)^{1/p}\}} \quad (92)$$

and so the bracketed term may be bounded

$$\left(\frac{1}{1 - \omega^{(\alpha-1)p/2}}\right)^{1/p} + \frac{1}{1 - \omega^{1/2}} \leq \frac{1}{1 - \omega^{1/2}} \cdot \frac{2}{\min\{1, (\alpha-1)^{1/p}\}} \quad (93)$$

and consequently

$$D_{\alpha,p,r}(f, f_\varphi) \leq C \cdot \|f\|_{M^r(\mathcal{U})} \cdot \frac{\varepsilon}{\tau_0^{1/2}} \cdot \frac{\tau_0^{\alpha/2}}{1 - \omega^{1/2}} \cdot \frac{1}{\min\{1, (\alpha-1)^{1/p}\}} = C \cdot \|f\|_{M^r(\mathcal{U})} \cdot \frac{\tau_0^{\alpha/2}}{1 - \omega^{1/2}} \cdot \Delta_{\alpha,p}(\varepsilon/\sqrt{\tau_0}), \quad (94)$$

as claimed. $\square$

4.2 Changes in projection angle

Suppose $f : \mathbb{R}^3 \rightarrow \mathbb{R}$, U is in $\text{SO}(3)$, and $f_U(x) = f(Ux)$. Let $e_z = (0, 0, 1)$, $v = Ue_z$, and $\theta \in [0, \pi)$ be the angle between e_z and v . Let $\mathcal{P}$ denote the tomographic projection operator onto the xy -plane:

$$(\mathcal{P}f)(x, y) = \int f(x, y, z) dz. \quad (95)$$

We then have the following result:

Theorem 4.7. *Let f be supported on the ball $B(0, R)$ of radius $R > 0$. Suppose $\tau_0 \geq (2R \sin(\theta/2))^2$. Then*

$$D_{\alpha,p,r}(\mathcal{P}f, \mathcal{P}f_U; \text{SO}(2)) \leq C \cdot \|f\|_{M^r} \cdot \frac{\tau_0^{\alpha/2}}{1 - \omega^{1/2}} \cdot \Delta_{\alpha,p}(2R \sin(\theta/2)/\sqrt{\tau_0}), \quad (96)$$

where $C > 0$ is a universal constant.

It is convenient to first state one elementary lemma, whose proof we sketch for the reader's convenience.

Lemma 4.8. *Suppose U is in $\text{SO}(3)$, and consider its domain to be $B(0, R)$, the ball of radius $R > 0$ centered at the origin. Then*

$$\varepsilon(U)^2 = \max_{x \in B(0, R)} |x - Ux|^2 = R^2(3 - \text{tr}(U)). \quad (97)$$

Proof of Lemma 4.8. Without loss of generality assume $R = 1$. Let v be the fixed axis of U , so that in an orthonormal basis starting with v , the matrix representation of U is of the form

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\phi) & -\sin(\phi) \\ 0 & \sin(\phi) & \cos(\phi) \end{bmatrix}, \quad (98)$$

where ϕ is the angle which U rotates points around v . Then U moves all points on the unit circle of the plane orthogonal to v the same distance, whose squared value is

$$\varepsilon(U)^2 = (1 - \cos(\phi))^2 + \sin(\phi)^2 = 2(1 - \cos(\phi)) = 3 - \text{tr}(U), \quad (99)$$

as claimed. $\square$

Proof of Theorem 4.7. Without loss of generality, suppose that v lies in the yz -plane. Then the rotation $U = U(\theta, \psi)$ may be described by the composition of a rotation by θ in the yz plane (bringing v to e_z) with a rotation in the xy -plane by another angle, ψ :

$$\begin{aligned} U(\theta, \psi) &= \begin{bmatrix} \cos(\psi) & -\sin(\psi) & 0 \\ \sin(\psi) & \cos(\psi) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\theta) & -\sin(\theta) \\ 0 & \sin(\theta) & \cos(\theta) \end{bmatrix} \\ &= \begin{bmatrix} \cos(\psi) & -\sin(\psi) \cos(\theta) & \sin(\psi) \sin(\theta) \\ \sin(\psi) & \cos(\psi) \cos(\theta) & -\cos(\psi) \sin(\theta) \\ 0 & \sin(\theta) & \cos(\theta) \end{bmatrix}. \end{aligned} \quad (100)$$

By Lemma 4.8,

$$\varepsilon(U(\theta, \psi))^2 = R^2(3 - \text{tr}(U(\theta, \psi))) = R^2(3 - \cos(\psi) - \cos(\psi)\cos(\theta) - \cos(\theta)). \quad (101)$$

Minimizing this over the choice of in-plane rotation ψ gives $\psi = 0$, and the corresponding squared distance is

$$\varepsilon(U(\theta, 0))^2 = R^2(2 - 2\cos(\theta)) = R^2(2 - 2(1 - 2\sin(\theta/2)^2)) = 4R^2\sin(\theta/2)^2. \quad (102)$$

Consequently, since ψ determines the in-plane rotation,

$$\begin{aligned} D_{\alpha,p,r}(\mathcal{P}f, \mathcal{P}f_{U(\theta,\psi)}; \text{SO}(2)) &= \min_{Q \in \text{SO}(2)} D_{\alpha,p,r}(\mathcal{P}f, \mathcal{P}f_{U(\theta,\psi)} \circ Q) \\ &= \min_{Q \in \text{SO}(2)} D_{\alpha,p,r}(\mathcal{P}f, \mathcal{P}f_{U(\theta,0)} \circ Q) \\ &\leq D_{\alpha,p,r}(\mathcal{P}f, \mathcal{P}f_{U(\theta,0)}). \end{aligned} \quad (103)$$

By Theorem 4.2, then, we have the bound

$$D_{\alpha,p,r}(\mathcal{P}f, \mathcal{P}f_{U(\theta,0)}) \leq C \cdot \|f\|_{M^r} \cdot \frac{\tau_0^{\alpha/2}}{1 - \omega^{1/2}} \cdot \Delta_{\alpha,p}(2R\sin(\theta/2)/\sqrt{\tau_0}), \quad (104)$$

as claimed. $\square$

Remark 6. Similar results bounding the distance between tomographic projections by an increasing function of the angle between projection directions are known for rotationally-invariant Wasserstein distance [38] and rotationally-invariant sliced Wasserstein distance [42]. Using the results from [22], a similar result can be easily deduced for the sliced Cramér metrics as well.

4.3 Convolutions

Images observed in scientific problems are often convolved with a linear filter induced from the measurement apparatus. In cryo-EM, for example, the projection images are filtered by the contrast transfer function [5, 44, 47, 11, 51]. It is therefore of interest to consider the effect of such linear filters on distances between the images. The following result is immediate:

Proposition 4.9. *Let $\alpha > 0$, $p \geq 1$, $r \geq 1$. Suppose f and g are in L^r , and w is in L^1 . Then*

$$D_{\alpha,p,r}(f * w, g * w) \leq \|w\|_{L^1} \cdot D_{\alpha,p,r}(f, g). \quad (105)$$

Proof. The proof is an immediate consequence of Young's inequality for convolutions. Indeed, for any τ ,

$$\|G_\tau * ((f * w) - (g * w))\|_{L^r} = \|w * G_\tau * (f - g)\|_{L^r} \leq \|w\|_{L^1} \|G_\tau * (f - g)\|_{L^r}, \quad (106)$$

from which the result follows. $\square$

Remark 7. Of course, the analogous result holds for any L^r distance, also due to Young's inequality. It is also known for other metrics, such as the sliced Cramér metric [54, 22].

5 Discretizations in 2D

We describe Fourier-based discretizations of the metrics in 2D. We only consider the case where $r = 2$, and denote the corresponding metric $D_{\alpha,p}(g, h)$.

Throughout, we let g and h be supported in $B(0, R_0)$, the ball of radius $R_0 > 0$ centered at the origin; and we let $L_0 = 2R_0$. We suppose $R > R_0$ is big enough so that $|(G_{\tau_0} * g)(t)|$ and $|(G_{\tau_0} * h)(t)|$ are negligibly small for $|t| \geq R$. More precisely, let $R = R_0 + \sqrt{\tau_0 \cdot \log(1/\epsilon)}$, where ϵ is a specified tolerance. We also let $L = 2R$. Let $t_j = -R + 2Rj/n$, $j = 0, \dots, n-1$. For simplicity, we will assume that n is even. We assume that we observe samples $g(t_{j_1}, t_{j_2})$ and $h(t_{j_1}, t_{j_2})$ of g and h on the Cartesian grid; note that if we only observe samples from the grid over $(-R_0, R_0) \times (-R_0, R_0)$, the additional samples over the full domain $(-R, R) \times (-R, R)$ can be obtained from zero-padding.

5.1 Discretization in 2D, without rotations

In this section, we let $f = g - h$; our goal is fast approximation of $D_{\alpha,p}(g, h) = \|f\|_{\Gamma_p^\alpha}$. For $j_1, j_2 = 0, \dots, n-1$, let $x[j_1, j_2] = f(t_{j_1}, t_{j_2}) = g(t_{j_1}, t_{j_2}) - h(t_{j_1}, t_{j_2})$ denote the samples of f on the Cartesian grid. We define the normalized DFT of x by

$$\hat{x}[k_1, k_2] = \frac{L^2}{n^2} \sum_{j_1=0}^{n-1} \sum_{j_2=0}^{n-1} x[j_1, j_2] e^{-2\pi i(k_1 t_{j_1} + k_2 t_{j_2})/L}, \quad (107)$$

so that, for $\max\{|k_1|, |k_2|\} \leq n/2$,

$$\hat{x}[k_1, k_2] \approx \widehat{f}(k_1/L, k_2/L). \quad (108)$$

Denote by $\widehat{G}_\tau(\xi_1, \xi_2)$ the Gaussian Fourier transform:

$$\widehat{G}_\tau(\xi_1, \xi_2) = e^{-\pi^2 \tau (\xi_1^2 + \xi_2^2)}, \quad (109)$$

and denote the approximate Fourier coefficients of $G_\tau * f$ by

$$\hat{x}_\tau[k_1, k_2] = \hat{x}[k_1, k_2] \widehat{G}_\tau(k_1/L, k_2/L), \quad (110)$$

so that

$$\hat{x}_\tau[k_1, k_2] \approx (\widehat{G_\tau * f})(k_1/L, k_2/L). \quad (111)$$

We then define the approximation to the L^2 norm at level τ by

$$S(\tau)^2 = \frac{1}{L^2} \sum_{k_1=-n/2}^{n/2-1} \sum_{k_2=-n/2}^{n/2-1} |\hat{x}_\tau[k_1, k_2]|^2. \quad (112)$$

Then

$$S(\tau)^2 \approx \|G_\tau * f\|_{L^2}^2. \quad (113)$$

Finally, we define the approximation to $\|f\|_{\Gamma_p^\alpha}$ by

$$\|x\|_{\gamma_p^\alpha} = \left(\sum_{\ell=0}^{M_n} \tau_\ell^{p\alpha/2} S(\tau_\ell)^p \right)^{1/p}, \quad (114)$$

for a specified $M_n = O(\log(n))$. Then $\|x\|_{\gamma_p^\alpha} \approx \|f\|_{\Gamma_p^\alpha} = D_{\alpha,p}(g, h)$.

Remark 8. Note that the calculation of $\|x\|_{\gamma_p^\alpha}$ can be performed with $O(n^2 \log(n))$ floating-point operations. Indeed, a 2D FFT computes all $\hat{x}[k_1, k_2]$ at cost $O(n^2 \log(n))$; and at each of the $M_n = O(\log(n))$ layers, n^2 multiplications are performed to evaluate all $\hat{x}_{\tau_\ell}[k_1, k_2]$ and evaluate the sums defining $S(\tau_\ell)^2$; then an additional $M_n = O(\log(n))$ operations are required to sum over all M_n levels.

We prove an error bound:

Theorem 5.1. *Suppose g and h are $C^\infty(\mathbb{R})$, and supported in $B(0, R_0)$. Let $f = g - h$. Fix $J > 1$, let $0 < \epsilon < n^{-J}$, and $R = R_0 + \sqrt{\tau_0} \cdot \log(1/\epsilon)$. Then using $M_n = \lfloor (2J/p\alpha) \log_{1/\omega}(n) \rfloor$ levels in the definition of $\|x\|_{\gamma_p^\alpha}$,*

$$\left| \|x\|_{\gamma_p^\alpha} - \|f\|_{\Gamma_p^\alpha} \right| = O\left(n^{-J}\right). \quad (115)$$

Remark 9. The constants implicit in the notation “ $O(n^{-J})$ ” may depend on $J, f, \alpha, \omega, \tau_0$, and p .

Proof of Theorem 5.1. It will first be convenient to record several elementary lemmas, some of whose proofs we provide for the reader's convenience.

Lemma 5.2. *Suppose and $|a - \tilde{a}| \leq \delta$ for some $\delta > 0$. Let $q \geq 1$. Then $||a|^q - |\tilde{a}|^q| \leq q \cdot (|a| + \delta)^{q-1} \cdot \delta$.*

Proof of Lemma 5.2. Let $h(x) = |x|^q$, so that $h'(x) = q|x|^{q-1} \text{sign}(x)$. By the Mean Value Theorem, there is some b between a and $\tilde{a}$ such that

$$||a|^q - |\tilde{a}|^q| = |h(a) - h(\tilde{a})| = |h'(b)| \cdot |a - \tilde{a}| = q|b|^{q-1} \cdot \delta < q \cdot (|a| + \delta)^{q-1} \cdot \delta, \quad (116)$$

where we have used that $|b| \leq \max\{|a|, |\tilde{a}|\} \leq |a| + \delta$.

□

Corollary 5.3. Suppose z and $\tilde{z}$ are complex, with $|z - \tilde{z}| \leq \delta$ for some $\delta > 0$. Then $||z|^2 - |\tilde{z}|^2| \leq 2\sqrt{2} \cdot |z| \cdot \delta + 4\delta^2$.

Proof of Corollary 5.3. If $z = a + ib$ and $\tilde{z} = \tilde{a} + i\tilde{b}$, then from Lemma 5.2, since $|a - \tilde{a}| \leq |z - \tilde{z}| \leq \delta$, therefore $|a^2 - \tilde{a}^2| \leq 2 \cdot (|a| + \delta) \cdot \delta$ and similarly, $|b^2 - \tilde{b}^2| \leq 2 \cdot (|b| + \delta) \cdot \delta$, and so

$$||z|^2 - |\tilde{z}|^2| = |a^2 + b^2 - (\tilde{a}^2 + \tilde{b}^2)| \leq |a^2 - \tilde{a}^2| + |b^2 - \tilde{b}^2| \leq 2 \cdot (|a| + \delta) \cdot \delta + 2 \cdot (|b| + \delta) \cdot \delta \leq 2\sqrt{2} \cdot |z| \cdot \delta + 4\delta^2, \quad (117)$$

as claimed. $\square$

Lemma 5.4. Suppose $a, \tilde{a} > 0$, and $|a - \tilde{a}| \leq \delta$ for some $\delta > 0$. Let $p \geq 1$. Then $|a^{1/p} - \tilde{a}^{1/p}| \leq \delta^{1/p}$.

Proof of Lemma 5.4. This is immediate from the inequality $(a+b)^{1/p} \leq a^{1/p} + b^{1/p}$ for $a, b > 0$. Indeed, we have $a \leq \tilde{a} + \delta$, and so $a^{1/p} \leq (\tilde{a} + \delta)^{1/p} \leq \tilde{a}^{1/p} + \delta^{1/p}$, hence $a^{1/p} - \tilde{a}^{1/p} \leq \delta^{1/p}$; and similarly $\tilde{a}^{1/p} - a^{1/p} \leq \delta^{1/p}$, completing the proof. $\square$

Lemma 5.5. Suppose $a, \tilde{a} > 0$, and $|a - \tilde{a}| < \delta$ for $0 < \delta < a/2$. Let $p \geq 1$. Then

$$|a^{1/p} - \tilde{a}^{1/p}| \leq (1/p)(a/2)^{1/p-1}\delta. \quad (118)$$

Proof. First, note that, since $a - \tilde{a} < \delta < a/2$, we have $\tilde{a} > a/2$. Let $q(t) = t^{1/p}$, for $t \geq 0$. Then $q'(t) = (1/p)t^{1/p-1}$, which is decreasing. By the Mean Value Theorem, there exists b between a and $\tilde{a}$ such that

$$|a^{1/p} - \tilde{a}^{1/p}| \leq (1/p)b^{1/p-1}|a - \tilde{a}| \leq (1/p)(a/2)^{1/p-1}\delta, \quad (119)$$

where we have used $b \geq a/2$. $\square$

Lemma 5.6. The function

$$q(\delta) = \frac{\delta}{1 - \delta} \log(1/\delta) \quad (120)$$

is increasing on $(0, 1)$, converges to 0 as $\delta \rightarrow 0^+$, and converges to 1 as $\delta \rightarrow 1^-$.

Proof of Lemma 5.6. The limits are obtained by l'Hôpital's rule. The derivative of the function is

$$q'(\delta) = \frac{\delta - 1 - \log(\delta)}{(\delta - 1)^2}, \quad (121)$$

which is positive over $0 < \delta < 1$ since $\log(\delta) < \delta - 1$. $\square$

Lemma 5.7. If $\tau < \tau_0$, then $G_\tau(x) \leq G_{\tau_0}(x)$ whenever

$$|x| \geq \sqrt{\tau_0 \tau \frac{\log(\tau_0/\tau)}{\tau_0 - \tau}}. \quad (122)$$

Furthermore, the right side decreases as τ decreases.

Proof of Lemma 5.7. The first part is immediate from rearranging the inequality $G_{\tau_0}(x) \geq G_\tau(x)$. The second part follows by rewriting the square of the right side as

$$\tau_0 \tau \frac{\log(\tau_0/\tau)}{\tau_0 - \tau} = \tau_0 \frac{\tau/\tau_0}{1 - \tau/\tau_0} \log(1/(\tau/\tau_0)) \quad (123)$$

and applying Lemma 5.6 with $\delta = \tau/\tau_0$. $\square$

We break the error analysis into several steps.

Step 1. For $\tau < \tau_0$, we will first bound the error

$$\left| \frac{1}{L^2} \sum_{(k_1, k_2) \in \mathbb{Z}^2} |\widehat{f}(k_1/L, k_2/L)|^2 \widehat{G}_\tau(k_1/L, k_2/L)^2 - \|G_\tau * f\|_{L^2}^2 \right|. \quad (124)$$

Let $f_s(x) = f(-x)$ for $x \in \mathbb{R}^2$, and let $H_\tau = (G_\tau * f) * (G_\tau * f_s)$. Note that

$$H_\tau(0, 0) = \|G_\tau * f\|_{L^2}^2. \quad (125)$$

Also note that H_τ has Fourier transform

$$\widehat{H}_\tau(\xi_1, \xi_2) = |\widehat{f}(\xi_1, \xi_2)|^2 \widehat{G}_\tau(\xi_1, \xi_2)^2. \quad (126)$$

By the Poisson summation formula (e.g. Theorem 8.32 in [15]),

$$\begin{aligned} \frac{1}{L^2} \sum_{k_1=-\infty}^{\infty} \sum_{k_2=-\infty}^{\infty} |\widehat{f}(k_1/L, k_2/L)|^2 \widehat{G}_\tau(k_1/L, k_2/L)^2 &= \frac{1}{L^2} \sum_{k_1=-\infty}^{\infty} \sum_{k_2=-\infty}^{\infty} \widehat{H}_\tau(k_1/L, k_2/L) \\ &= \sum_{j_1=-\infty}^{\infty} \sum_{j_2=-\infty}^{\infty} H_\tau(j_1 L, j_2 L) \\ &= H_\tau(0, 0) + \sum_{(j_1, j_2) \neq (0, 0)} H_\tau(j_1 L, j_2 L). \\ &= \|G_\tau * f\|_{L^2}^2 + \sum_{(j_1, j_2) \neq (0, 0)} H_\tau(j_1 L, j_2 L). \end{aligned} \quad (127)$$

We now need to bound the remainder term. Let $F = f * f_s$, and note that, since f is assumed to be supported in $B(0, R_0)$, F is supported in $B(0, L_0)$, where $L_0 = 2R_0$. Note too that $G_\tau * G_\tau = G_{2\tau}$, and so

$$H_\tau = F * G_{2\tau}. \quad (128)$$

Therefore,

$$\begin{aligned} \sum_{j \in \mathbb{Z}^2 \setminus (0, 0)} H_\tau(jL) &= \sum_{j \in \mathbb{Z}^2 \setminus (0, 0)} \int_{B(0, L_0)} F(x) G_{2\tau}(x - jL) dx \\ &\leq \int_{B(0, L_0)} |F(x)| \sum_{j \in \mathbb{Z}^2 \setminus (0, 0)} G_{2\tau}(x - jL) dx. \end{aligned} \quad (129)$$

We bound the series appearing in the integral uniformly, as follows. First, since the ℓ_∞ ball of radius L_0 is bigger than the ℓ_2 ball of radius L_0 , it is enough to bound

$$\sup_{|x|_\infty \leq L_0} \sum_{j \in \mathbb{Z}^2 \setminus (0, 0)} G_{2\tau}(x - jL). \quad (130)$$

Suppose $j = (j_1, j_2) \neq (0, 0)$, and without loss of generality suppose that $j_1 > 0$. Then the maximum possible value of $G_{2\tau}(x - jL)$ for $|x|_\infty \leq L_0$ is achieved when $x_1 = L_0$, in which case

$$|jL - x|^2 = |j_1 L - x_1|^2 + |j_2 L - x_2|^2 \geq |j_1 L - L_0|^2. \quad (131)$$

Now, for $\tau \leq \tau_0$, say $\tau = \tau_0 \cdot \delta$ with $0 < \delta < 1$, by Lemma 5.7, then $G_{2\tau}(y) \leq G_{2\tau_0}(y)$ when

$$\begin{aligned} |y|^2 &\geq 2\tau_0 \tau \frac{\log(\tau_0/\tau)}{\tau_0 - \tau} \\ &= 2\tau_0^2 \delta \frac{\log(1/\delta)}{\tau_0 - \tau_0 \delta} \\ &= 2\tau_0 \frac{\delta}{1 - \delta} \log(1/\delta). \end{aligned} \quad (132)$$

From Lemma 5.6, this last quantity is bounded above by $2\tau_0$; hence, so long as $|y| \geq \sqrt{2\tau_0}$, we have $G_{2\tau}(y) \leq G_{2\tau_0}(y)$. With $x_1 = L_0$, we have

$$|jL - x|^2 \geq |j_1 L - L_0|^2 \geq |L - L_0|^2 = 4|R - R_0|^2 = 4\tau_0 \cdot \log(1/\epsilon) \geq 4\tau_0, \quad (133)$$

it then follows that for this choice of x , $G_{2\tau}(|jL - x|) \leq G_{2\tau_0}(|jL - x|)$. Hence, it is enough to prove an upper bound for $\tau = \tau_0$.

To that end, we have

$$\begin{aligned}
G_{2\tau_0}(x - jL) &\leq \frac{1}{2\tau_0} e^{-(j_1 L - x_1)^2 / (2\tau_0)} \\
&\leq \frac{1}{2\tau_0} e^{-(j_1 L - L_0)^2 / (2\tau_0)} \\
&= \frac{1}{2\tau_0} e^{-(j_1 L - j_1 L_0 + (j_1 - 1)L_0)^2 / (2\tau_0)} \\
&\leq \frac{1}{2\tau_0} e^{-(j_1 L - j_1 L_0)^2 / (2\tau_0)} \\
&= \frac{1}{2\tau_0} e^{-j_1^2 (L - L_0)^2 / (2\tau_0)} \\
&= \frac{1}{2\tau_0} e^{-j_1^2 4\tau_0 \log(1/\epsilon) / (2\tau_0)} \\
&= \frac{1}{2\tau_0} \epsilon^{2j_1^2}.
\end{aligned} \tag{134}$$

The same holds, of course, if $j_1 < 0$. Similarly, if $j_2 \neq 0$, then

$$G_{2\tau_0}(x - jL) \leq \frac{1}{2\tau_0} \epsilon^{2j_2^2}. \tag{135}$$

And if both $j_1 \neq 0$ and $j_2 \neq 0$, then similar reasoning shows that

$$G_{2\tau_0}(x - jL) \leq \frac{1}{2\tau_0} \epsilon^{2(j_1^2 + j_2^2)}. \tag{136}$$

Hence

$$\begin{aligned}
\sum_{j \in \mathbb{Z}^2 \setminus (0,0)} G_{2\tau}(x - jL) &\leq \frac{1}{2\tau_0} \sum_{(j_1, j_2) \neq (0,0)} \epsilon^{2(j_1^2 + j_2^2)} \\
&= \frac{1}{2\tau_0} \left(\sum_{j_1 \neq 0} \epsilon^{2j_1^2} + \sum_{j_2 \neq 0} \epsilon^{2j_2^2} + \sum_{j_1 \neq 0, j_2 \neq 0} \epsilon^{2j_1^2} \epsilon^{2j_2^2} \right) \\
&= \frac{1}{\tau_0} \sum_{m \neq 0} \epsilon^{2m^2} + \frac{1}{2\tau_0} \left(\sum_{m \neq 0} \epsilon^{2m^2} \right)^2 \\
&\leq \frac{C}{\tau_0} \epsilon^2,
\end{aligned} \tag{137}$$

for a universal constant $C > 0$.

Therefore,

$$\begin{aligned}
\sum_{j \in \mathbb{Z}^2 \setminus (0,0)} H_\tau(Lj) &\leq \int_{B(0, L_0)} |F(x)| \sum_{j \in \mathbb{Z}^2 \setminus (0,0)} G_{2\tau}(x - jL) dx \\
&\leq \frac{C}{\tau_0} \|F\|_{L^1} \epsilon^2 \\
&\leq \frac{C}{\tau_0} \|f\|_{L^1}^2 \epsilon^2,
\end{aligned} \tag{138}$$

where in the last inequality we have used that $\|F\|_{L^1} = \|f * f_s\|_{L^1} \leq \|f\|_{L^1} \|f_s\|_{L^1} = \|f\|_{L^1}^2$. We have therefore shown that

$$\left| \frac{1}{L^2} \sum_{k_1 = -\infty}^{\infty} \sum_{k_2 = -\infty}^{\infty} |\widehat{f}(k_1/L, k_2/L)|^2 \widehat{G}_\tau(k_1/L, k_2/L)^2 - \|G_\tau * f\|_{L^2}^2 \right| \leq \frac{C}{\tau_0} \|f\|_{L^1}^2 \epsilon^2. \tag{139}$$

Step 2. We now bound the error from truncating the infinite sum. Because f is C^∞ , for any $J > 0$, $|\widehat{f}(\xi)| = O(|\xi|^{-(J+2)})$, and so $|\widehat{f}(k/L)|^2 = O(L^{2J+2}/|k|^{2J+2})$. Furthermore, $\|\widehat{G}_\tau\|_\infty \leq 1$ for all τ . Note too that for any $m \geq 1$, there are $8m$

$k \in \mathbb{Z}^2$ with $|k|_\infty = m$. It then follows that the remainder term can be bounded:

$$\begin{aligned}
& \left| \frac{1}{L^2} \sum_{k \in \mathbb{Z}^2} |\widehat{f}(k_1/L, k_2/L)|^2 \widehat{G}_\tau(k_1/L, k_2/L)^2 - \frac{1}{L^2} \sum_{|k|_\infty \leq n/2} |\widehat{f}(k_1/L, k_2/L)|^2 \widehat{G}_\tau(k_1/L, k_2/L)^2 \right| \\
&= \frac{1}{L^2} \sum_{|k|_\infty > n/2} |\widehat{f}(k_1/L, k_2/L)|^2 \widehat{G}_\tau(k_1/L, k_2/L)^2 \\
&\leq C \cdot L^{2J} \sum_{|k|_\infty > n/2} \frac{1}{|k|^{2J+2}} \\
&= C \cdot L^{2J} \sum_{m=n/2+1}^{\infty} \sum_{|k|_\infty = m} \frac{1}{m^{2J+2}} \\
&= C \cdot L^{2J} \sum_{m=n/2+1}^{\infty} \frac{8}{m^{2J+1}} \\
&\leq C \cdot L^{2J} \int_{n/2}^{\infty} \frac{1}{r^{2J+1}} dr \\
&= C \cdot \frac{L^{2J}}{2J} \cdot \frac{1}{(n/2)^{2J}} \\
&= C \cdot \frac{(2L)^{2J}}{J} \cdot \frac{1}{n^{2J}}, \tag{140}
\end{aligned}$$

i.e. we have shown the bound

$$\left| \frac{1}{L^2} \sum_{k \in \mathbb{Z}^2} |\widehat{f}(k_1/L, k_2/L)|^2 \widehat{G}_\tau(k_1/L, k_2/L)^2 - \frac{1}{L^2} \sum_{|k|_\infty \leq n/2} |\widehat{f}(k_1/L, k_2/L)|^2 \widehat{G}_\tau(k_1/L, k_2/L)^2 \right| = O(n^{-2J}). \tag{141}$$

Step 3. Finally, we account for the error in approximating each $\widehat{f}(k_1/L, k_2/L)$ by $\widehat{x}[k_1, k_2]$. Because f is C^∞ , a standard error bound (see e.g. [22]) says that for $\max\{|k_1|, |k_2|\} \leq n/(2L)$,

$$|\widehat{x}[k_1, k_2] - \widehat{f}(k_1/L, k_2/L)| = O(n^{-(2J+2)}), \tag{142}$$

and so from Corollary 5.3,

$$\left| |\widehat{x}[k_1, k_2]|^2 - |\widehat{f}(k_1/L, k_2/L)|^2 \right| = O(n^{-(2J+2)}). \tag{143}$$

Consequently, since $\widehat{G}_\tau(k_1/L, k_2/L) \leq 1$, summing the errors over all n^2 terms gives the error bound

$$\left| \sum_{k_1=-n/2}^{n/2-1} \sum_{k_2=-n/2}^{n/2-1} \left(|\widehat{x}[k_1, k_2]|^2 \widehat{G}_\tau(k_1/L, k_2/L)^2 - |\widehat{f}(k_1/L, k_2/L)|^2 \widehat{G}_\tau(k_1/L, k_2/L)^2 \right) \right| = O(n^{-2J}). \tag{144}$$

Step 4. Putting all the bounds together,

$$\left| \|G_\tau * f\|_{L^2}^2 - \frac{1}{L^2} \sum_{k_1=-n/2}^{n/2-1} \sum_{k_2=-n/2}^{n/2-1} |\widehat{x}[k_1, k_2]|^2 \widehat{G}_\tau(k_1/L, k_2/L)^2 \right| = O(n^{-2J} + \epsilon^2) = O(n^{-2J}). \tag{145}$$

Taking square roots and employing Lemma 5.4,

$$\left| \|G_\tau * f\|_{L^2} - \left(\frac{1}{L^2} \sum_{k_1=-n/2}^{n/2-1} \sum_{k_2=-n/2}^{n/2-1} |\widehat{x}[k_1, k_2]|^2 \widehat{G}_\tau(k_1/L, k_2/L)^2 \right)^{1/2} \right| = O(n^{-J}), \tag{146}$$

i.e.

$$\left| \|G_\tau * f\|_{L^2} - S(\tau_\ell) \right| = O(n^{-J}). \tag{147}$$

Letting δ denote the $O(n^{-J})$ error, for n sufficiently big, $\delta \leq \|f\|_{L^2}$, and so by Lemma 5.2,

$$\left| \|G_\tau * f\|_{L^2}^p - S(\tau)^p \right| \leq C \cdot p \cdot (\|G_\tau * f\|_{L^2} + \delta)^{p-1} \cdot \delta \leq C \cdot p \cdot (2\|f\|_{L^2})^{p-1} \cdot n^{-J} = O(n^{-J}). \tag{148}$$

Importantly, the constant hidden in the “ $O(n^{-J})$ ” does not depend on τ .

Step 5. We now bound the error for the entire series defining the metric. As we have shown, there is a $C > 0$ such that for all τ ,

$$\|G_\tau * f\|_{L^2}^p - S(\tau)^p \leq C \cdot n^{-J}. \quad (149)$$

Recall that $M_n = \lfloor (2J/p\alpha) \log_{1/\omega}(n) \rfloor$. Therefore,

$$\begin{aligned} \left| \sum_{k=0}^{M_n} \tau_k^{p\alpha/2} (S(\tau_k)^p - \|G_{\tau_k} * f\|_{L^2}^p) \right| &\leq C \cdot n^{-J} \cdot \tau_0^{p\alpha/2} \sum_{k=0}^{M_n} \omega^{kp\alpha/2} \\ &\leq C \cdot n^{-J} \cdot \tau_0^{p\alpha/2} \frac{1}{1 - \omega^{p\alpha/2}}. \end{aligned} \quad (150)$$

Regarding the remainder,

$$\begin{aligned} \sum_{k=M_n+1}^{\infty} \tau_k^{p\alpha/2} \|G_{\tau_k} * f\|_{L^2}^p &\leq \|f\|_{L^2}^p \sum_{k=M_n+1}^{\infty} \tau_k^{p\alpha/2} \\ &\leq \|f\|_{L^2}^p \tau_0^{p\alpha/2} \sum_{k=M_n+1}^{\infty} \omega^{kp\alpha/2} \\ &= \|f\|_{L^2}^p \tau_0^{p\alpha/2} \frac{\omega^{(M_n+1)p\alpha/2}}{1 - \omega^{p\alpha/2}}. \end{aligned} \quad (151)$$

So, since $M_n = \lfloor (2J/p\alpha) \log_{1/\omega}(n) \rfloor$, therefore

$$\omega^{M_n+1} \leq n^{-2J/p\alpha}, \quad (152)$$

and the bound becomes

$$\sum_{k=M_n+1}^{\infty} \tau_k^{p\alpha/2} \|G_{\tau_k} * f\|_{L^2}^p \leq \|f\|_{L^2}^p \tau_0^{p\alpha/2} \frac{n^{-J}}{1 - \omega^{p\alpha/2}} \quad (153)$$

Putting the bounds together,

$$\left| \|x\|_{\gamma_p^\alpha}^p - \|f\|_{\Gamma_p^\alpha}^p \right| = O(n^{-J}). \quad (154)$$

Assume $\|f\|_{\Gamma_p^\alpha} > 0$ (otherwise the result is trivial, since $f \equiv 0$ and hence $x \equiv 0$). For n sufficiently big, $\|x\|_{\gamma_p^\alpha}^p \geq \|f\|_{\Gamma_p^\alpha}^p/2$; consequently, by Lemma 5.5,

$$\left| \|x\|_{\gamma_p^\alpha} - \|f\|_{\Gamma_p^\alpha} \right| = O(n^{-J}), \quad (155)$$

concluding the proof. $\square$

5.2 Discretization in 2D, with in-plane rotations

Next we consider the discretization of the minimum distance $D_{\alpha,p}(g, h; \text{SO}(2))$. As before, we suppose access to samples $g(t_{j1}, t_{j2})$ and $h(t_{j1}, t_{j2})$ of g and h on a Cartesian grid.

By abuse of notation, we denote by $\widehat{G}_\tau(r)$ the Fourier transform of the Gaussian G_τ at any (ξ_1, ξ_2) with $r = \sqrt{\xi_1^2 + \xi_2^2}$. Also abusing notation, denote by $\widehat{g}(r, \theta)$ and $\widehat{h}(r, \theta)$ the functions $\widehat{g}$ and $\widehat{h}$ in polar coordinates.

For $0 < \tau \leq \tau_0$, define the inner product function Q_τ by

$$Q_\tau(\psi; g, h) = \int_0^{2\pi} \int_0^\infty \widehat{g}(r, \theta) \widehat{h}(r, \theta - \psi)^* \widehat{G}_\tau(r)^2 r dr d\theta, \quad (156)$$

where here $*$ denotes complex conjugation. Then $Q_\tau(\psi; g, h)$ is the inner product between $G_\tau * g$ and the rotation of $G_\tau * h$ by ψ .

We introduce several key definitions and observations, which we will be able to convert into an efficient algorithm:

1. For $r > 0$, write the Fourier expansions

$$\widehat{g}(r, \theta) = \sum_j a_j(r) e^{ij\theta}, \quad (157)$$

and

$$\widehat{h}(r, \theta) = \sum_j b_j(r) e^{ij\theta}, \quad (158)$$

for $0 \leq \theta < 2\pi$.

2. For each fixed r and ψ ,

$$\int_0^{2\pi} \widehat{g}(r, \theta) \widehat{h}(r, \theta - \psi)^* d\theta = \sum_j a_j(r) b_j(r)^* e^{ij\psi}. \quad (159)$$

3. The Fourier expansion of Q_τ may then be written as follows:

$$\begin{aligned} Q_\tau(\psi; g, h) &= \int_0^\infty \int_0^{2\pi} \widehat{g}(r, \theta) \widehat{h}(r, \theta - \psi)^* \widehat{G}_\tau(r)^2 d\theta r dr \\ &= \int_0^\infty \left(\sum_j a_j(r) b_j(r)^* e^{ij\psi} \right) \widehat{G}_\tau(r)^2 r dr \\ &= \sum_j \left(\int_0^\infty a_j(r) b_j(r)^* \widehat{G}_\tau(r)^2 r dr \right) e^{ij\psi}. \end{aligned} \quad (160)$$

That is, Q_τ has Fourier coefficients

$$\widehat{Q}_\tau[j] = \int_0^\infty a_j(r) b_j(r)^* \widehat{G}_\tau(r)^2 r dr. \quad (161)$$

4. Let $D_{\alpha,p}(g, h; \psi)$ denote the distance between g and the rotation of h by ψ . Then

$$D_{\alpha,p}(g, h; \psi)^p = \sum_{k=0}^\infty \tau_k^{p\alpha/2} (\|G_{\tau_k} * g\|_{L^2}^2 + \|G_{\tau_k} * h\|_{L^2}^2 - 2Q_{\tau_k}(\psi; g, h))^{p/2}. \quad (162)$$

We now describe the algorithm. Let $R = n/(2L)$; this is the maximum frequency resolvable from the observed samples. Take integers N_r , N_θ , and N_ψ , where $N_\psi \geq N_\theta$. Let $r_k = R \cdot k/N_r$, $k = 0, \dots, N_r$, be the radial values; let $\theta_\ell = 2\pi\ell/N_\theta$, $\ell = 0, \dots, N_\theta - 1$, be the interpolation angles, which we assume without loss of generality to be even; and let $\psi_m = 2\pi m/N_\psi$, $m = 0, \dots, N_\psi - 1$, be the angles at which we will approximate the $Q_{\tau_\ell}(\psi_m)$. Following the discussion in Section 2.2 of [2], we take $N_\theta = O(n)$, and $N_r = O(n)$. We can take N_ψ substantially larger than N_θ without incurring additional asymptotic cost (though the constant obviously grows); see Remark 11 below.

1. Using a non-uniform fast Fourier transform, or NUFFT [3, 4, 12, 13, 17, 14], evaluate all sums

$$\widetilde{g}[k, \ell] = \frac{L^2}{n^2} \sum_{j_1=1}^n \sum_{j_2=1}^n g(t_{j_1}, t_{j_2}) e^{-2\pi i(t_{j_1} r_k \cos(\theta_\ell) + t_{j_2} r_k \sin(\theta_\ell))} \quad (163)$$

at cost $O(n^2 \log(n))$. These values approximate $\widehat{g}(r_k, \theta_\ell)$, the 2D Fourier transform of g at the points $(r_k \cos(\theta_\ell), r_k \sin(\theta_\ell))$. We do the same for h , with values $\widehat{h}[k, \ell] \approx \widehat{h}(r_k, \theta_\ell)$.

Remark 10. In our implementation, we use the FINUFFT from the Flatiron Institute [3, 4].

2. For each $k = 0, \dots, N_r$, evaluate the discrete Fourier coefficients $\widetilde{a}[k, j]$ of $\widetilde{g}[k, \cdot]$, and the discrete Fourier coefficients $\widetilde{b}[k, j]$ of $\widetilde{h}[k, \cdot]$, $j = -N_\theta/2, \dots, N_\theta/2$. Then $\widetilde{a}[k, j] \approx a_j(r_k)$ and $\widetilde{b}[k, j] \approx b_j(r_k)$, for $k = 0, \dots, N_r$ and $j = -N_\theta/2, \dots, N_\theta/2$. Using an FFT, for each k the cost of computing all coefficients $\widetilde{a}[k, j]$ and $\widetilde{b}[k, j]$, $|j| \leq N_\theta/2$, is $O(N_\theta \log(N_\theta)) = O(n \log(n))$, and so the total cost over all $N_r = O(n)$ radii r_k is $O(n^2 \log(n))$.
3. For each ℓ and j evaluate the approximate integral

$$\widetilde{Q}[j, \ell] = \frac{R}{N_r} \sum_{k=0}^{N_r} \widetilde{a}[k, j] \widetilde{b}[k, j]^* \widehat{G}_{\tau_k}(r_k)^2 r_k \approx \widehat{Q}_{\tau_\ell}[j]. \quad (164)$$

For each j , the sum may be evaluated at cost $O(N_r) = O(n)$; and since there are $N_\theta = O(n)$ Fourier coefficients j and $M_n = O(\log(n))$ levels ℓ , the total cost is $O(n^2 \log(n))$.

Having computed approximations to the Fourier coefficients $\widehat{Q}_{\tau_\ell}[j]$, for each $\ell = 0, \dots, M_n$, we can approximate the inner products $Q_{\tau_\ell}(\psi_m; g, h)$, $m = 0, \dots, N_\psi - 1$, at cost $O(N_\psi \log(N_\psi))$, by first zero-padding the Fourier coefficients $\widehat{Q}_{\tau_\ell}[j]$ and then computing an IFFT. Repeating this computation for all M_n levels incurs a total cost of $O(N_\psi \log(N_\psi) \cdot M_n)$.

Remark 11. Because the cost of this step is $O(N_\psi \log(N_\psi) \cdot M_n)$ and $M_n = O(\log(n))$, we can take N_ψ to be substantially larger than n (nearly on the order of $O(n^2)$) without increasing the asymptotic cost $O(n^2 \log(n))$ of the algorithm, allowing for a finer resolution alignment.

4. All M_n norms $\|G_{\tau_\ell} * g\|_{L^2}$ and $\|G_{\tau_\ell} * h\|_{L^2}$, $\ell = 0, \dots, M$, can be approximated at cost $O(n^2 \log(n))$: first compute the 2D FFTs of g and h at cost $O(n^2 \log(n))$, and then, at each level $\ell = 0, \dots, M_n$, multiply the DFTs of g and h by the Fourier coefficients of G_{τ_ℓ} at total cost $O(n^2 \log(n))$; and sum the coefficients for each level at cost $O(n^2 \log(n))$. Alternatively, the norms can be computed using the samples of $\hat{g}$ and $\hat{h}$ already computed on the polar grid.

We can then approximate all N_ψ rotated distances by using the formula

$$D_{\alpha,p}(g, h; \psi_m)^p = \sum_{k=0}^{M_n} \tau_k^{p\alpha/2} (\|G_{\tau_k} * g\|_{L^2}^2 + \|G_{\tau_k} * h\|_{L^2}^2 - 2Q_{\tau_k}(\psi_m; g, h))^{p/2}, \quad (165)$$

at an additional cost of $O(N_\psi \log(n))$. The entire cost of the calculation of all rotated distances, therefore, is $O(n^2 \log(n))$.

6 Robustness to noise in 2D

Suppose the random variables $Z[j_1, j_2]$, $0 \leq j_1, j_2 \leq n-1$ are independent, zero mean sub-Gaussians with respective sub-Gaussian norms $\sigma[j_1, j_2] = \|Z[j_1, j_2]\|_{\psi_2}$; and suppose

$$\frac{1}{n} \sum_{j=0}^{n-1} \sigma[j_1, j_2]^2 \leq \sigma^2 \quad (166)$$

for all n , where $\sigma > 0$. Define the random variable

$$\Pi_n \equiv \|Z\|_{\gamma_p^\alpha} = \left(\sum_{\ell=0}^M \tau_\ell^{p\alpha/2} S(\tau_\ell)^p \right)^{1/p}. \quad (167)$$

We will bound the sub-Gaussian norm of Π_n :

Theorem 6.1. *The random variable $\Pi_n = \|Z\|_{\gamma_p^\alpha}$ has sub-Gaussian norm bounded as follows:*

$$\|\Pi_n\|_{\psi_2} \leq C \cdot \frac{\tau_0^\alpha}{1 - \omega^{1/2}} \cdot L \cdot \sqrt{p} \cdot \sigma \cdot \Delta_{\alpha,p}(L/n\sqrt{\tau_0}), \quad (168)$$

where $C > 0$ is a universal constant.

Remark 12. In particular, when $\alpha > 1$, the sub-Gaussian norm decays like $1/n$, which is the inverse square root decay (n being the square root of the total number of observations) we would expect from averaging independent random variables. However, when $0 < \alpha \leq 1$, the rate of decay is worse: at $\alpha = 1$, the decay is like $(1 + \log(n))^{1/p}/n$, whereas when $0 < \alpha < 1$, it decays at the even slower rate of $1/n^\alpha$. This is because with smaller values, the weights $\tau_k^{p\alpha/2}$ decay slower, and hence place relatively more weight on the higher-level Gaussian filters, which do not filter the noise as aggressively.

Remark 13. By Proposition 2.6.6 of [48], we have the concentration inequality

$$P(\|Z\|_{\gamma_p^\alpha} > t) \leq 2e^{-t^2/\|\Pi_n\|_{\psi_2}^2}. \quad (169)$$

It follows immediately (using Borel-Cantelli) that $\|Z\|_{\gamma_p^\alpha}$ converges to 0 almost surely as $n \rightarrow \infty$. Furthermore, we also have the bound on the expected value of $\|Z\|_{\gamma_p^\alpha}$:

$$\mathbb{E} \left[\|Z\|_{\gamma_p^\alpha} \right] \leq C \cdot \frac{\tau_0^\alpha}{1 - \omega^{1/2}} \cdot L \cdot \sqrt{p} \cdot \sigma \cdot \Delta_{\alpha,p}(L/n\sqrt{\tau_0}), \quad (170)$$

which also follows immediately from Proposition 2.6.6 of [48], with a possibly different universal constant C .

We now prove Theorem 6.1.

Proof of Theorem 6.1. We have

$$\widehat{Z}[k_1, k_2] = \frac{L^2}{n^2} \sum_{j_1=0}^{n-1} \sum_{j_2=0}^{n-1} Z[j_1, j_2] e^{-2\pi i(k_1 t_{j_1} + k_2 t_{j_2})/L}, \quad (171)$$

The $Z[j_1, j_2]$ are independent sub-Gaussians with mean zero, and by Proposition 2.7.1 in [48] there is a universal $C > 0$ such that

$$\left\| \widehat{Z}[k_1, k_2] \right\|_{\psi_2}^2 \leq C \frac{L^4}{n^4} \sum_{j_1=0}^{n-1} \sum_{j_2=0}^{n-1} \|Z[j_1, j_2]\|_{\psi_2}^2 = C \frac{L^4}{n^2} \left(\frac{1}{n^2} \sum_{j_1=0}^{n-1} \sum_{j_2=0}^{n-1} \sigma_j^2 \right) \leq CL^4 \frac{\sigma^2}{n^2}. \quad (172)$$

(Note that while Proposition 2.7.1 in [48] is stated for real-valued random variables, it is not hard to extend to complex-valued random variables.)

Therefore, the sub-Gaussian norm of

$$\widehat{Z}_\tau[k_1, k_2] = \widehat{Z}[k_1, k_2] \widehat{G}_\tau(k_1/L, k_2/L) \quad (173)$$

is similarly bounded above:

$$\left\| \widehat{Z}_\tau[k_1, k_2] \right\|_{\psi_2}^2 = CL^4 \frac{\sigma^2}{n^2} \widehat{G}_\tau(k_1/L, k_2/L)^2. \quad (174)$$

For $\tau \leq \tau_0$, we now bound the sub-Gaussian norm of the random variable

$$S(\tau)^2 = \frac{1}{L^2} \sum_{k_1=-n/2}^{n/2-1} \sum_{k_2=-n/2}^{n/2-1} |\widehat{Z}_\tau[k_1, k_2]|^2. \quad (175)$$

Since $\|S(\tau)\|_{\psi_1}^2 = \|S(\tau)\|_{\psi_2}^2$, and since $\|\cdot\|_{\psi_1}$ is a norm, therefore,

$$\begin{aligned} \|S(\tau)\|_{\psi_2}^2 &= \|S(\tau)\|_{\psi_1}^2 \\ &\leq \frac{1}{L^2} \sum_{k_1=-n/2}^{n/2-1} \sum_{k_2=-n/2}^{n/2-1} \left\| |\widehat{Z}_\tau[k_1, k_2]|^2 \right\|_{\psi_1} \\ &= \frac{1}{L^2} \sum_{k_1=-n/2}^{n/2-1} \sum_{k_2=-n/2}^{n/2-1} \left\| |\widehat{Z}_\tau[k_1, k_2]| \right\|_{\psi_2}^2 \\ &\leq \frac{1}{L^2} \sum_{k_1=-n/2}^{n/2-1} \sum_{k_2=-n/2}^{n/2-1} CL^4 \frac{\sigma^2}{n^2} \widehat{G}_\tau(k_1/L, k_2/L)^2 \\ &= C\sigma^2 \frac{L^2}{n^2} \sum_{k_1=-n/2}^{n/2-1} \sum_{k_2=-n/2}^{n/2-1} \widehat{G}_\tau(k_1/L, k_2/L)^2. \end{aligned} \quad (176)$$

To bound this sum, we factor

$$\widehat{G}_\tau(k_1/L, k_2/L)^2 = e^{-2\pi^2\tau(k_1/L)^2} e^{-2\pi^2\tau(k_2/L)^2}, \quad (177)$$

which allows us to write the bound in (176) as

$$\|S(\tau)\|_{\psi_2}^2 \leq CL^4 \frac{\sigma^2}{n^2} \left(\frac{1}{L} \sum_{k=-n/2}^{n/2-1} e^{-2\pi^2\tau(k/L)^2} \right)^2. \quad (178)$$

To bound this: we write

$$\begin{aligned} \frac{1}{L} \sum_{k=-n/2}^{n/2-1} e^{-2\pi^2\tau(k/L)^2} &\leq \frac{1}{L} + \frac{2}{L} \sum_{k=1}^{n/2} e^{-2\pi^2\tau(k/L)^2} \\ &\leq \frac{1}{L} + \frac{2}{L} \sum_{k=1}^{\infty} e^{-2\pi^2\tau(k/L)^2} \\ &\leq \frac{1}{L} + 2 \sum_{k=1}^{\infty} \int_{(k-1)/L}^{k/L} e^{-2\pi^2\tau\xi^2} d\xi \\ &= \frac{1}{L} + 2 \int_0^{\infty} e^{-2\pi^2\tau\xi^2} d\xi \\ &= \frac{1}{L} + \frac{2}{\sqrt{\tau}} \int_0^{\infty} e^{-2\pi^2\nu^2} d\nu. \end{aligned} \quad (179)$$

Therefore,

$$\|S(\tau)\|_{\psi_2}^2 \leq CL^4 \frac{\sigma^2}{n^2} \left(\frac{1}{L} + \frac{2}{\sqrt{\tau}} \int_0^{\infty} e^{-2\pi^2\nu^2} d\nu \right)^2 \leq CL^4 \frac{\sigma^2}{n^2} \frac{1}{\tau}, \quad (180)$$

for a possibly different constant $C > 0$, and where we have used that $\sqrt{\tau} \leq \sqrt{\tau_0} \leq L$ (indeed, $L = L_0 + 2\sqrt{\tau_0 \log(1/\epsilon)}$ for small $\epsilon > 0$).

Of course, we also have the more trivial bound

$$\frac{1}{L} \sum_{k=-n/2}^{n/2-1} e^{-2\pi^2 \tau (k/L)^2} \leq \frac{n}{L}, \quad (181)$$

from which it follows that

$$\|S(\tau)\|_{\psi_2}^2 \leq CL^4 \frac{\sigma^2}{n^2} \left(\frac{1}{L} \sum_{k=-n/2}^{n/2-1} e^{-2\pi^2 \tau (k/L)^2} \right)^2 \leq CL^4 \frac{\sigma^2}{n^2} \frac{n^2}{L^2} = CL^2 \sigma^2. \quad (182)$$

To bound the sub-Gaussian norm of Π_n itself, we now have the following lemma, whose proof we provide for the convenience of the reader.

Lemma 6.2. *Suppose $X_1, \dots, X_m$ are sub-Gaussian random variables, $p \geq 1$, and $X = (\sum_{i=1}^m |X_i|^p)^{1/p}$. Then X is sub-Gaussian, with*

$$\|X\|_{\psi_2} \leq C\sqrt{p} \left(\sum_{i=1}^m \|X_i\|_{\psi_2}^p \right)^{1/p}, \quad (183)$$

where $C > 0$ is a universal constant.

Proof of Lemma 6.2. From Proposition 2.6.1 in [48], the sub-Gaussian norm $\|X\|_{\psi_2}$ is equivalent to the norm $\|X\|_{\psi_2}^{(1)} = \sup_{r \geq 1} r^{-1/2} (\mathbb{E}[|X|^r])^{1/r}$, in the sense that $\|X\|_{\psi_2}^{(1)} / \|X\|_{\psi_2}$ is bounded above and below by universal constants. Fix $r \geq 1$. We consider two cases.

Case 1: $r \leq p$. First, as just remarked, by Proposition 2.6.1 in [48] there is a universal $C > 0$ such that

$$\mathbb{E}[|X_i|^p] \leq C^p p^{p/2} \|X_i\|_{\psi_2}^p. \quad (184)$$

Because $r/p \leq 1$, by Jensen's inequality we have

$$\begin{aligned} \mathbb{E}[X^r] &\leq \mathbb{E} \left[\left(\sum_{i=1}^m |X_i|^p \right)^{r/p} \right] \\ &\leq \left[\mathbb{E} \left(\sum_{i=1}^m |X_i|^p \right) \right]^{r/p} \\ &= \left[\sum_{i=1}^m \mathbb{E}[|X_i|^p] \right]^{r/p} \\ &\leq \left[\sum_{i=1}^m C^p p^{p/2} \|X_i\|_{\psi_2}^p \right]^{r/p} \\ &= C^r p^{r/2} \left[\sum_{i=1}^m \|X_i\|_{\psi_2}^p \right]^{r/p}, \end{aligned} \quad (185)$$

and taking r -th roots, dividing by $\sqrt{r}$, and using that $r \geq 1$ and hence $r^{-1/2} \leq 1$, we have

$$r^{-1/2} (\mathbb{E}[|X|^r])^{1/r} \leq r^{-1/2} C\sqrt{p} \left[\sum_{i=1}^m \|X_i\|_{\psi_2}^p \right]^{1/p} \leq C\sqrt{p} \left[\sum_{i=1}^m \|X_i\|_{\psi_2}^p \right]^{1/p}, \quad (186)$$

which is the desired bound.

Case 2: $r > p$. For clarity, let $Y_i = |X_i|^p$, and let $q = r/p > 1$. Again, using Proposition 2.6.1 in [48], $(\mathbb{E}[|X_i|^r])^{1/r} \leq C\sqrt{r} \|X_i\|_{\psi_2}$ for universal $C > 0$.

Using, for instance, Minkowski's inequality for integrals (e.g. Theorem 6.19 of [15]), we have the following inequality:

$$\begin{aligned}
(\mathbb{E}[X^r])^{1/r} &= \left(\mathbb{E} \left[\left(\sum_{i=1}^n |X_i|^p \right)^{r/p} \right] \right)^{1/r} \\
&= \left(\left(\mathbb{E} \left[\left(\sum_{i=1}^n Y_i \right)^q \right] \right)^{1/q} \right)^{q/r} \\
&\leq \left(\sum_{i=1}^n (\mathbb{E}[Y_i^q])^{1/q} \right)^{q/r} \\
&= \left(\sum_{i=1}^n (\mathbb{E}[|X_i|^r])^{p/r} \right)^{1/p} \\
&\leq \left(\sum_{i=1}^n C^p r^{p/2} \|X_i\|_{\psi_2}^p \right)^{1/p} \\
&= Cr^{1/2} \left(\sum_{i=1}^n \|X_i\|_{\psi_2}^p \right)^{1/p}, \tag{187}
\end{aligned}$$

and so

$$r^{-1/2}(\mathbb{E}[X^r])^{1/r} \leq C \left(\sum_{i=1}^n \|X_i\|_{\psi_2}^p \right)^{1/p}, \tag{188}$$

which is the desired bound. $\square$

Using Lemma 6.2 we immediately get the bound

$$\|\Pi_n\|_{\psi_2} = \left\| \left(\sum_{\ell=0}^M \tau_\ell^{p\alpha/2} S(\tau_\ell)^p \right)^{1/p} \right\|_{\psi_2} \leq C\sqrt{p} \left(\sum_{\ell=0}^M \tau_\ell^{p\alpha/2} \|S(\tau_\ell)\|_{\psi_2}^p \right)^{1/p} \leq C\sqrt{p} \left(\sum_{\ell=0}^\infty \tau_\ell^{p\alpha/2} \|S(\tau_\ell)\|_{\psi_2}^p \right)^{1/p}, \tag{189}$$

for universal $C > 0$.

Let

$$N = \lfloor 2 \log_{1/\omega}(n\sqrt{\tau_0}/L) \rfloor. \tag{190}$$

We break the proof into the three regimes of α appearing in the definition of $\Delta_{\alpha,p}$.

Case 1: $0 < \alpha < 1$. First, break the series into two parts:

$$\sum_{\ell=0}^\infty \tau_\ell^{p\alpha/2} \|S(\tau_\ell)\|_{\psi_2}^p = \sum_{\ell=0}^N \tau_\ell^{p\alpha/2} \|S(\tau_\ell)\|_{\psi_2}^p + \sum_{\ell=N+1}^\infty \tau_\ell^{p\alpha/2} \|S(\tau_\ell)\|_{\psi_2}^p. \tag{191}$$

For the first sum, we use (180), which implies

$$\|S(\tau_\ell)\|_{\psi_2}^p \leq C^{p/2} L^{2p} \frac{\sigma^p}{n^p \tau_\ell^{p/2}}, \tag{192}$$

and hence

$$\begin{aligned}
\sum_{\ell=0}^N \tau_{\ell}^{p\alpha/2} \|S(\tau_{\ell})\|_{\psi_2}^p &\leq C^{p/2} L^{2p} \frac{\sigma^p}{n^p} \sum_{\ell=0}^N \tau_{\ell}^{p\alpha/2} \frac{1}{\tau_{\ell}^{p/2}} \\
&\leq C^{p/2} L^{2p} \frac{\sigma^p}{n^p} \sum_{\ell=0}^N \tau_{\ell}^{p(\alpha-1)/2} \\
&= C^{p/2} L^{2p} \frac{\sigma^p}{n^p} \tau_0^{p(\alpha-1)/2} \sum_{\ell=0}^N \omega^{\ell p(\alpha-1)/2} \\
&= C^{p/2} L^{2p} \frac{\sigma^p}{n^p} \tau_0^{p(\alpha-1)/2} \frac{\omega^{Np(\alpha-1)/2}}{1 - \omega^{p(1-\alpha)/2}} \\
&\leq C^{p/2} L^{2p} \frac{\sigma^p}{n^p} \tau_0^{p(\alpha-1)/2} \frac{n^{p(1-\alpha)} L^{p(\alpha-1)} \tau_0^{p(1-\alpha)/2}}{1 - \omega^{p(1-\alpha)/2}} \\
&\leq C^{p/2} L^{p(\alpha+1)} \frac{\sigma^p}{n^{p\alpha}} \frac{1}{1 - \omega^{p(1-\alpha)/2}}. \\
&\leq C^{p/2} L^{p(\alpha+1)} \tau_0^{-p\alpha/2} \frac{\sigma^p}{n^{p\alpha}} \frac{\tau_0^{p\alpha/2}}{1 - \omega^{p/2}} \cdot \frac{1}{1 - \alpha}.
\end{aligned} \tag{193}$$

Using the bound (182), or equivalently, $\|S(\tau)\|_{\psi_2}^p \leq C^{p/2} L^p \sigma^p$, the second sum can be bounded by

$$\begin{aligned}
\sum_{\ell=N+1}^{\infty} \tau_{\ell}^{p\alpha/2} \|S(\tau_{\ell})\|_{\psi_2}^p &\leq C^{p/2} L^p \sigma^p \sum_{\ell=N+1}^{\infty} \tau_{\ell}^{p\alpha/2} \\
&= C^{p/2} L^p \sigma^p \tau_0^{p\alpha/2} \sum_{\ell=N+1}^{\infty} \omega^{\ell p\alpha/2} \\
&\leq C^{p/2} L^p \sigma^p \tau_0^{p\alpha/2} \frac{\omega^{(N+1)p\alpha/2}}{1 - \omega^{p\alpha/2}} \\
&\leq C^{p/2} L^p \sigma^p \tau_0^{p\alpha/2} \frac{n^{-p\alpha} L^{p\alpha} \tau_0^{-p\alpha/2}}{1 - \omega^{p\alpha/2}} \\
&= C^{p/2} L^{p(\alpha+1)} \tau_0^{-p\alpha/2} \frac{\sigma^p}{n^{p\alpha}} \frac{\tau_0^{p\alpha/2}}{1 - \omega^{p\alpha/2}} \\
&\leq C^{p/2} L^{p(\alpha+1)} \tau_0^{-p\alpha/2} \frac{\sigma^p}{n^{p\alpha}} \frac{\tau_0^{p\alpha/2}}{1 - \omega^{p/2}} \cdot \frac{1}{\alpha}.
\end{aligned} \tag{194}$$

Adding (193) and (194)

$$\begin{aligned}
p^{-p/2} \|\Pi_n\|_{\psi_2}^p &\leq C^{p/2} L^{p(\alpha+1)} \tau_0^{-p\alpha/2} \frac{\sigma^p}{n^{p\alpha}} \frac{\tau_0^{p\alpha/2}}{1 - \omega^{p/2}} \cdot \frac{1}{1 - \alpha} + C^{p/2} L^{p(\alpha+1)} \tau_0^{-p\alpha/2} \frac{\sigma^p}{n^{p\alpha}} \frac{\tau_0^{p\alpha/2}}{1 - \omega^{p/2}} \cdot \frac{1}{\alpha} \\
&= C^{p/2} L^{p(\alpha+1)} \tau_0^{-p\alpha/2} \frac{\sigma^p}{n^{p\alpha}} \frac{\tau_0^{p\alpha/2}}{1 - \omega^{p/2}} \left(\frac{1}{1 - \alpha} + \frac{1}{\alpha} \right) \\
&= C^{p/2} L^{p(\alpha+1)} \tau_0^{-p\alpha/2} \frac{\sigma^p}{n^{p\alpha}} \frac{\tau_0^{p\alpha/2}}{1 - \omega^{p/2}} \cdot \frac{1}{\alpha - \alpha^2}.
\end{aligned} \tag{195}$$

Taking p -th roots, and for a different universal $C > 0$,

$$\begin{aligned}
p^{-1/2} \|\Pi_n\|_{\psi_2} &\leq C \cdot L \cdot \tau_0^{\alpha/2} \left(\frac{1}{1 - \omega^{p/2}} \right)^{1/p} \cdot L^{\alpha} \cdot \tau_0^{-\alpha/2} \cdot \frac{\sigma}{n^{\alpha}} \cdot \frac{1}{(\alpha - \alpha^2)^{1/p}} \\
&\leq C \cdot L \cdot \sigma \cdot \frac{\tau_0^{\alpha/2}}{1 - \omega^{1/2}} \cdot \Delta_{\alpha,p}(L/n\sqrt{\tau_0}),
\end{aligned} \tag{196}$$

which is the desired bound.

Case 2: $\alpha = 1$. In this case, we again split the sum into two parts:

$$\sum_{\ell=0}^{\infty} \tau_{\ell}^{p/2} \|S(\tau_{\ell})\|_{\psi_2}^p = \sum_{\ell=0}^N \tau_{\ell}^{p/2} \|S(\tau_{\ell})\|_{\psi_2}^p + \sum_{\ell=N+1}^{\infty} \tau_{\ell}^{p/2} \|S(\tau_{\ell})\|_{\psi_2}^p. \tag{197}$$

Again using (180), or equivalently

$$\|S(\tau_\ell)\|_{\psi_2}^p \leq C^{p/2} L^{2p} \frac{\sigma^p}{n^p} \frac{1}{\tau_\ell^{p/2}}, \quad (198)$$

and using the inequality (80) which states that $\log(1/t) \geq 1 - t$ for $0 < t \leq 1$, the first sum can be bounded by

$$\begin{aligned} \sum_{\ell=0}^N \tau_\ell^{p/2} \|S(\tau_\ell)\|_{\psi_2}^p &\leq C^{p/2} L^{2p} \frac{\sigma^p}{n^p} \sum_{\ell=0}^N \tau_\ell^{p/2} \frac{1}{\tau_\ell^{p/2}} \\ &\leq C^{p/2} L^{2p} \frac{\sigma^p}{n^p} N \\ &\leq C^{p/2} L^{2p} \frac{\sigma^p}{n^p} \log_{1/\omega}(n\sqrt{\tau_0}/L) \\ &= C^{p/2} L^{2p} \frac{\sigma^p}{n^p} \frac{\log(n\sqrt{\tau_0}/L)}{\log(1/\omega)} \\ &= C^{p/2} L^{2p} \frac{\sigma^p}{n^p} \frac{(p/2) \log(n\sqrt{\tau_0}/L)}{\log(1/\omega^{p/2})} \\ &= C^{p/2} L^{2p} \frac{\sigma^p}{n^p} \frac{(p/2) \log(n\sqrt{\tau_0}/L)}{1 - \omega^{p/2}} \end{aligned} \quad (199)$$

And again using the bound $\|S(\tau)\|_{\psi_2}^2 \leq CL^2 \sigma^2$ from (182), the second sum can be bounded by

$$\begin{aligned} \sum_{\ell=N+1}^{\infty} \tau_\ell^{p/2} \|S(\tau_\ell)\|_{\psi_2}^p &\leq C^{p/2} L^p \sigma^p \tau_0^{p/2} \sum_{\ell=N+1}^{\infty} \omega^{\ell p/2} \\ &= C^{p/2} L^p \sigma^p \tau_0^{p/2} \frac{\omega^{(N+1)p/2}}{1 - \omega^{p/2}} \\ &\leq C^{p/2} L^p \sigma^p \tau_0^{p/2} \frac{\tau_0^{-p/2} n^{-p} L^p}{1 - \omega^{p/2}} \\ &= C^{p/2} L^{2p} \frac{\sigma^p}{n^p} \frac{1}{1 - \omega^{p/2}} \end{aligned} \quad (200)$$

Therefore, adding (199) and (200),

$$\begin{aligned} p^{-p/2} \|\Pi_n\|_{\psi_2}^p &\leq C^{p/2} L^{2p} \frac{\sigma^p}{n^p} \frac{(p/2) \log(n\sqrt{\tau_0}/L)}{1 - \omega^{p/2}} + C^{p/2} L^{2p} \frac{\sigma^p}{n^p} \frac{1}{1 - \omega^{p/2}} \\ &= C^{p/2} \cdot L^{2p} \cdot \tau_0^{-p\alpha/2} \cdot \frac{\sigma^p}{n^p} \cdot \frac{\tau_0^{p\alpha/2}}{1 - \omega^{p/2}} \cdot ((p/2) \log(n\sqrt{\tau_0}/L) + 1) \end{aligned} \quad (201)$$

and taking p -th roots, using that $p^{1/p}$ is bounded, and with a different universal $C > 0$, gives

$$\begin{aligned} p^{-1/2} \|\Pi_n\|_{\psi_2} &\leq C \cdot L^2 \cdot \tau_0^{-\alpha/2} \cdot \frac{\sigma}{n} \cdot \tau_0^{\alpha/2} \cdot \left(\frac{1}{1 - \omega^{p/2}} \right)^{1/p} \cdot (\log(n\sqrt{\tau_0}/L) + 1)^{1/p} \\ &\leq C \cdot L^2 \cdot \tau_0^{-\alpha/2} \cdot \frac{\sigma}{n} \cdot \tau_0^{\alpha/2} \cdot \frac{1}{1 - \omega^{1/2}} \cdot (\log(n\sqrt{\tau_0}/L) + 1)^{1/p} \\ &= C \cdot L \cdot \sigma \cdot \frac{\tau_0^{\alpha/2}}{1 - \omega^{1/2}} \cdot \Delta_{1,p}(L/n\sqrt{\tau_0}), \end{aligned} \quad (202)$$

as desired.

Case 3: $\alpha > 1$. We use (180), or equivalently

$$\|S(\tau_\ell)\|_{\psi_2}^p \leq C^{p/2} L^{2p} \frac{\sigma^p}{n^p} \frac{1}{\tau_\ell^{p/2}}, \quad (203)$$

and write

$$\begin{aligned}
\sum_{\ell=0}^{\infty} \tau_{\ell}^{p\alpha/2} \|S(\tau_{\ell})\|_{\psi_2}^p &\leq C^{p/2} L^{2p} \frac{\sigma^p}{n^p} \sum_{\ell=0}^{\infty} \tau_{\ell}^{p\alpha/2} \frac{1}{\tau_{\ell}^{p/2}} \\
&= C^{p/2} L^{2p} \frac{\sigma^p}{n^p} \sum_{\ell=0}^{\infty} \tau_{\ell}^{p(\alpha-1)/2} \\
&= C^{p/2} L^{2p} \frac{\sigma^p}{n^p} \tau_0^{p(\alpha-1)/2} \sum_{\ell=0}^{\infty} \omega^{\ell p(\alpha-1)/2} \\
&= C^{p/2} L^{2p} \frac{\sigma^p}{n^p} \frac{\tau_0^{p(\alpha-1)/2}}{1 - \omega^{p(\alpha-1)/2}}.
\end{aligned} \tag{204}$$

Using (92) from Case 3 of the proof of Theorem 4.2, we have

$$\left(\frac{1}{1 - \omega^{(\alpha-1)p/2}} \right)^{1/p} \leq \frac{1}{1 - \omega^{1/2}} \cdot \frac{1}{\min\{1, (\alpha-1)^{1/p}\}} \tag{205}$$

and so it follows that, for a possibly different universal $C > 0$,

$$\begin{aligned}
p^{-1/2} \|\Pi_n\|_{\psi_2} &\leq C \cdot \frac{\sigma}{n} \cdot L^2 \frac{\tau_0^{(\alpha-1)/2}}{(1 - \omega^{(\alpha-1)p/2})^{1/p}} \\
&\leq C \cdot \frac{\tau_0^{\alpha/2}}{1 - \omega^{1/2}} \cdot \tau_0^{-1/2} \cdot \frac{\sigma}{n} \cdot L^2 \cdot \frac{1}{\min\{1, (\alpha-1)^{1/p}\}} \\
&= C \cdot L \cdot \sigma \cdot \frac{\tau_0^{\alpha/2}}{1 - \omega^{1/2}} \cdot \Delta_{\alpha,p}(L/n\sqrt{\tau_0}),
\end{aligned} \tag{206}$$

which completes the proof of Theorem 6.1. □

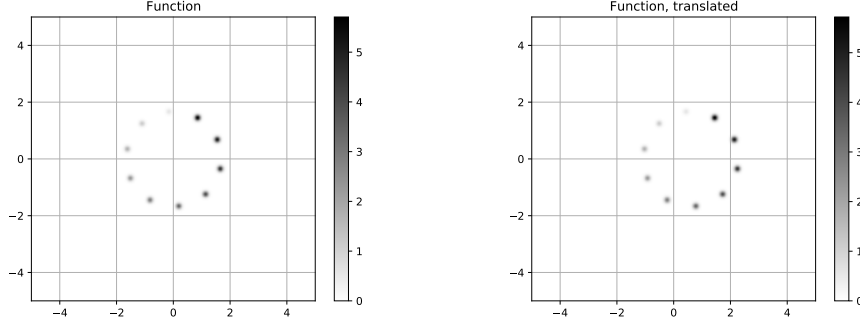


Figure 1: The function f described in Section 7.1, and its translation by 0.6 to the right.

7 Numerical experiments

We illustrate the performance of the rotationally-invariant metrics $D_{\alpha,p}(f, g; \text{SO}(2))$ (with $r = 2$), on several numerical examples. We compare to the rotationally-invariant Euclidean, sliced 2-Cramér, and sliced 2-Wasserstein distances. For simplicity, we take $p = 1$, and denote the metric by D_{α} . For our implementations of the rotationally-invariant sliced Wasserstein and sliced Cramér distances, we use similar implementations as those described in [42], though we make use of the observation from Remark 11 and perform the minimization over a finer grid than the algorithm described in [42]. The code for rotationally-invariant sliced Cramér and sliced Wasserstein distances may be found at github.com/wleeb/SlicedCramer. The code for the multiscale distances $D_{\alpha,p}$ (with $r = 2$), including the rotationally-invariant versions, may be found at github.com/wleeb/Gaussian-Kernel-Metrics.

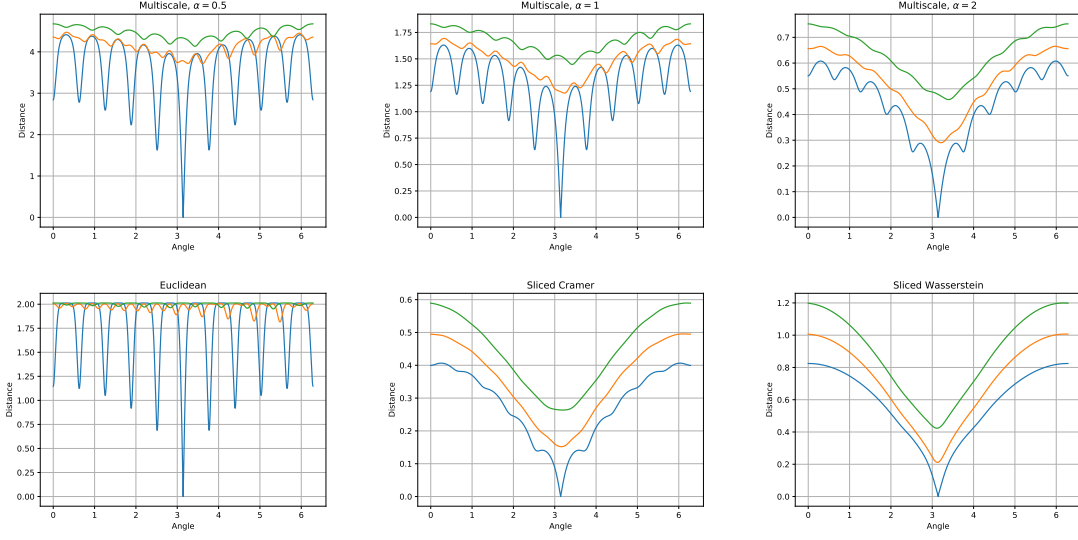


Figure 2: Distances for the experiment described in Section 7.1. From top to bottom, left to right: $D_{1/2}$, D_1 , D_2 , Euclidean, sliced Cramér, and sliced Wasserstein. The blue lines are the distances between the function and itself (rotated by π radians so the minimum value of 0 occurs in the middle of the plot); the orange lines are the distances between the function and its translation by 0.3; the green lines are the distances between the function and its translation by 0.6.

7.1 Translations and rotations

We consider the function f , shown in Figure 1, consisting of 10 Gaussians arranged equispaced around a circle, of varying heights. Each Gaussian is centered at distance $5/3$ from the origin, with scale parameter $1/100$, and heights that grow linearly starting from $1/55$ and ending at $10/55$ (the normalization is to ensure the total mass is 1). We evaluate distances between the function shown and all of its rotations, using the fast algorithm described in Section 5.2. We also compute distances between the original function and its translations to the right by 0.3 and 0.6, respectively. We do this for the distances D_α , $\alpha = 0.5, 1, 2$ (with $p = 1$) as well as sliced Wasserstein, sliced Cramér, and Euclidean distance. The distances are shown in Figure 3. As α grows, the behavior of the distances D_α are smoother functions of the rotation angle, similar to the sliced Wasserstein and sliced Cramér distances; whereas for smaller α , D_α contains the contours of the Euclidean distance, while still maintaining an overall envelope similar to that of the sliced Wasserstein and sliced Cramér distances.

7.2 Noise and rotations

We compare the rotations between the two functions shown in Figure 1, where white noise of varying strengths has been added to the rotated function. In this experiment, we do not include the sliced Wasserstein distance, as it is not designed to handle additive noise. The functions are sampled on grids of size 1592×1592 , which is substantially greater than is needed to resolve the underlying noiseless functions. The variances are, respectively, 0.5, 1.0, and 1.5. As can be seen, the noise drastically inflates the Euclidean distances; because the sample size is so large, the effects on the other metrics are comparatively smaller, and get smaller with larger α , consistent with Theorem 6.1.

7.3 Projections

We consider a function f in 3D, which is a convex combination of 24 Gaussian functions arranged in a 4-by-3-by-2 grid, each with scale parameter $1/100$, and heights proportional to $\sqrt{i^2 + j^2 + k^2}$, where (i, j, k) is the index specifying each Gaussian, with $i = 1, 2$, $j = 1, 2, 3$, and $k = 1, 2, 3, 4$. (The heights are normalized so the total mass is 1, making f a probability density.) The function f is centered at the origin, with the centers of the Gaussians equally spaced along each dimension, ranging from $-5/8$ to $5/8$.

We consider two 2D projections of f , whose axes of projection have angle $\pi/2$; these two functions, which we call g and h , are shown in the top row of Figure 4. Denote by u_g the axis along which f is projected to generate g , and similarly denote u_h .

In the experiment, for a given angle θ , we randomly generate two sets of 500 projections of f , which we denote $S_\theta(g)$ and $S_\theta(h)$, to which we add independent Gaussian noise of variance σ^2 . Some example projections without noise ($\sigma = 0$) are shown in the bottom two rows of Figure 4. All the projections in $S_\theta(g)$ (respectively $S_\theta(h)$) are taken at projection axes that lie within θ of u_g (respectively u_h).

For each projection in $S_\theta(g)$ and $S_\theta(h)$, we compute its distances to both g and h , using the rotationally-invariant versions of the D_2 , Euclidean, sliced Cramér, and, in the $\sigma = 0$ case, sliced Wasserstein metrics, and for each distance

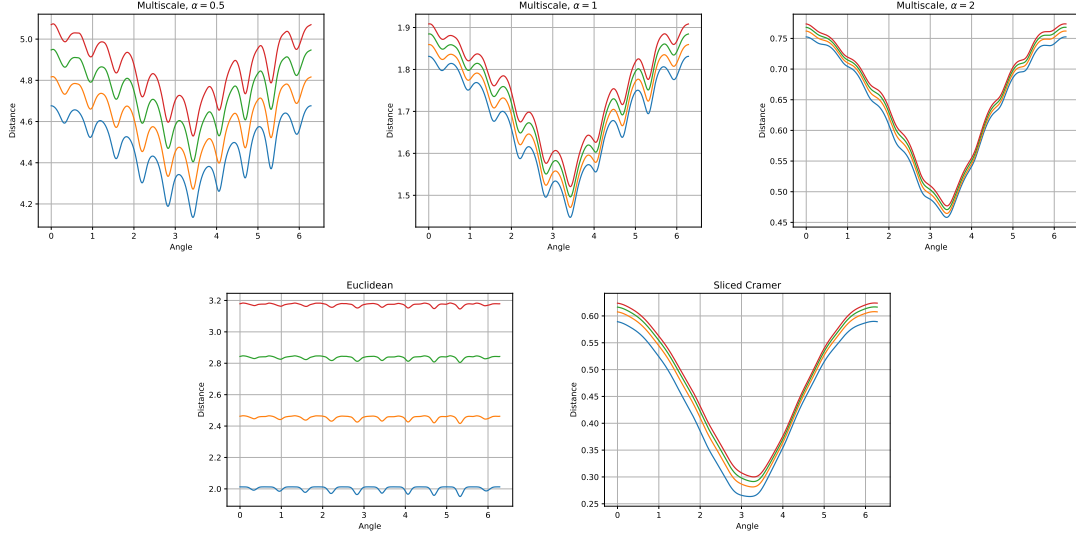


Figure 3: Distances for the experiment described in Section 7.2. From top to bottom, left to right: $D_{1/2}$, D_1 , D_2 , Euclidean, and sliced Cramér. The blue lines are the distances between the function and itself (rotated by π radians so the minimum value of 0 occurs in the middle of the plot); the orange lines are the distances between the functions with noise of maximum variance 0.5; the green lines are the distances between the functions with noise of maximum variance 1; the red lines are the distances between the functions with noise of maximum variance 1.5.

assign that projection to its nearest center (either g or h). For the metric D_2 , we take $\tau_0 = 25/16$.

In Figure 5, we plot the fraction of incorrectly classified noiseless ($\sigma = 0$) projections, as the angle θ grows, for all four distances. In Figures 6 and 7 we plot the fraction of incorrectly classified projections, for $\sigma = 0.1$ and $\sigma = 0.2$, respectively, as the angle θ grows; in these latter two plots, we exclude the sliced Wasserstein metrics, which are not designed to handle additive noise. As one would expect, all errors grow with θ , since the projections differ more as the angle between them changes.

Interestingly, in all cases the metric D_2 maintains the smallest errors. In the noiseless case, the errors are close to the errors of sliced Cramér and sliced Wasserstein, all three of which are considerably smaller than the errors of Euclidean distance. With non-zero noise, the metric D_2 performs substantially better than both the Euclidean and sliced Cramér distances. This is suggestive, though by no means conclusive, that the multiscale metric may perform well on the class averaging problem in cryo-EM [33, 18, 6, 38, 55]; however, additional experimentation is needed that goes beyond the scope of the current paper and will be pursued in future work.

8 Conclusion

We have shown that the metrics $D_{\alpha,p,r}$ are stable to geometric deformations, and that their discretizations are stable under additive noise. We have also introduced rotationally-invariant versions of $D_{\alpha,p,2}$, and we have shown that they may be computed efficiently from finite samples. Our numerical results indicate favorable performance compared to the sliced Wasserstein, sliced Cramér, and Euclidean distances.

There are several open questions that we intend to address in future work. Perhaps most straightforward is minimizing the distance over all rigid motions, not just over $SO(2)$. While minimizing over translations alone is fairly straightforward (more so than over rotations as we have done in the present work, as it does not require interpolating to a polar grid), simultaneous minimization over all rigid motions is a more challenging task, though we are optimistic that the methods in [36, 37] may be applicable. Second is the question of how to select the parameters α and p ; in particular, larger α puts more weight on the low-frequency components of the inputs, which makes the metric more stable but may erase critical details in the higher frequencies. On the other hand, larger p approaches the ∞ norm across scales, which selects a single scale to discriminate the two images, whereas values of p close to 1 incorporate all scales simultaneously. We also intend to extend this work to comparing volumes; Wasserstein-type distances have been proposed for this task in the context of cryo-EM [45, 53], and so it is reasonable to generalize the multiscale metrics studied herein to apply the same class of problems. Finally, Theorem 4.7 and the experiments in Section 7.3 suggest that in certain cases the rotationally-invariant metrics $D_{\alpha,p}$ may be good choices for clustering noisy tomographic images, a task that occurs in the class averaging problem in cryo-EM [33, 18, 6, 38, 55]; in future work we intend to design and test an end-to-end clustering algorithm for this task.

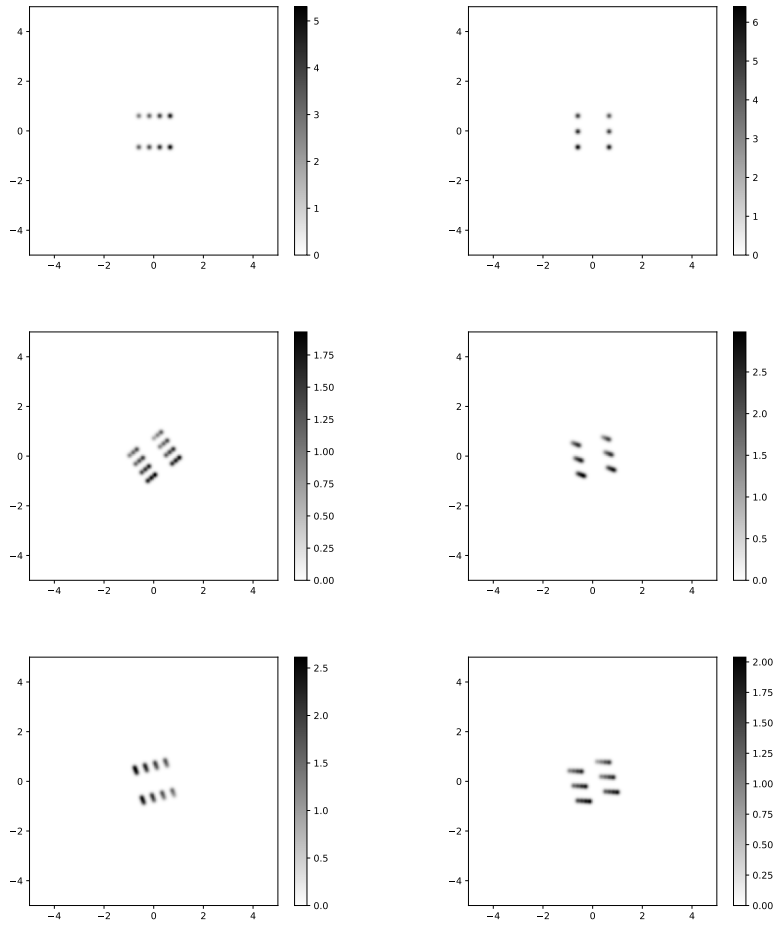


Figure 4: Projections of f from the experiment described in Section 7.3. Left column: the projection g , and two projections of f within angle θ of u_g . Right column: projection h , and two projections of f within angle θ of u_h . Here, $\theta = 3\pi/20$, and the projections are shown without noise.

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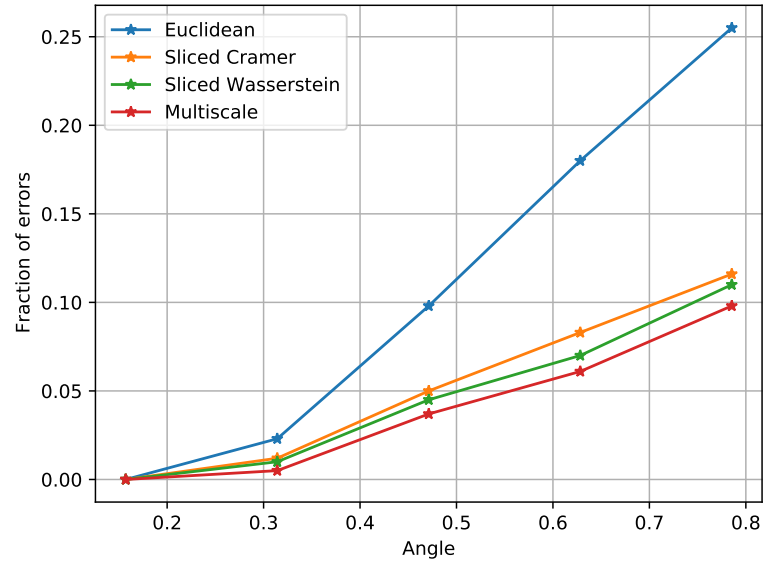


Figure 5: Error plot from the experiment described in Section 7.3.

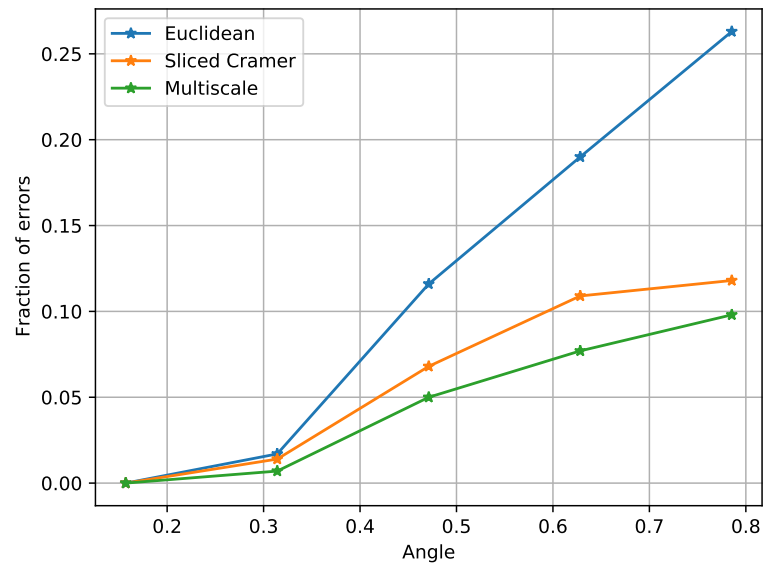


Figure 6: Error plot from the experiment described in Section 7.3, with noise of standard deviation 0.1.

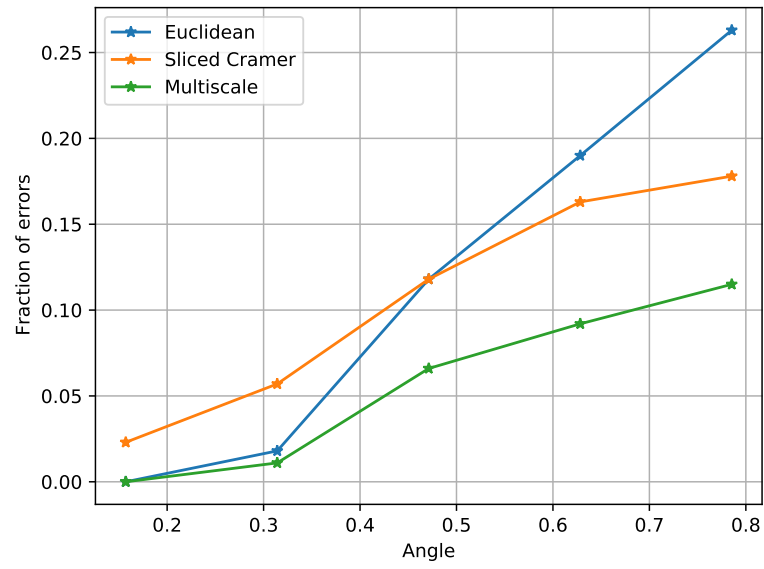


Figure 7: Error plot from the experiment described in Section 7.3, with noise of standard deviation 0.2.

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